

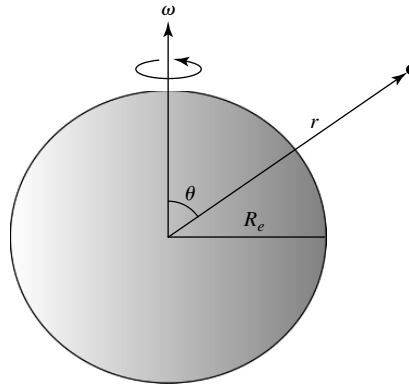


FIGURE 21.3 The angle θ is defined from the rotation axis for the Legendre polynomial expansion of the gravitational potential.

TABLE 21.2 Gravitational Moments and Moment-of-Inertia Ratios of the Giant Planets. R_e is the equatorial radius of the giant planet. (Data from Table 1 of Guillot, *Annu. Rev. Earth Planet. Sci.*, 33, 493, 2005.)

Moments	Jupiter	Saturn
J_2	$(1.4697 \pm 0.0001) \times 10^{-2}$	$(1.6332 \pm 0.0010) \times 10^{-2}$
J_4	$-(5.84 \pm 0.05) \times 10^{-4}$	$-(9.19 \pm 0.40) \times 10^{-4}$
J_6	$(0.31 \pm 0.20) \times 10^{-4}$	$(1.04 \pm 0.50) \times 10^{-4}$
I/MR_e^2	0.258	0.220

Moments	Uranus	Neptune
J_2	$(0.35160 \pm 0.00032) \times 10^{-2}$	$(0.3539 \pm 0.0010) \times 10^{-2}$
J_4	$-(0.354 \pm 0.041) \times 10^{-4}$	$-(0.28 \pm 0.22) \times 10^{-4}$
I/MR_e^2	0.230	0.241

shape. For example, J_2 is related to the planet's oblateness and to its **moment of inertia**.⁴ The J_4 and J_6 terms are more sensitive to the mass distribution in the outer regions of the planet, particularly the equatorial bulge, because the terms have stronger dependence on R_e . Because density is more dependent on temperature near the surface of the planet than it is in the deep interior where the gas tends to be degenerate, J_4 and J_6 also measure the planet's thermal structure. Gravitational moments for the giant planets are given in Table 21.2.

Example 21.1.1. The first three higher-order gravitational moments for Jupiter are given in Table 21.2. As a result, the associated expansion terms in Eq. (21.1) have the values shown in Fig. 21.4. The contribution of oblateness to the gravitational potential near the equator

continued

⁴The moment of inertia has already been discussed for the cases of Earth and the Moon (see Problems 20.12 and 20.13, respectively), and will be explored again for Jupiter (e.g., Problem 21.3).

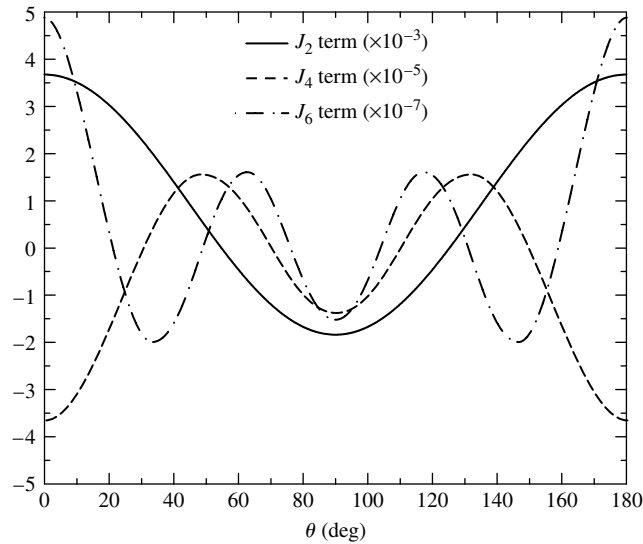


FIGURE 21.4 The first three higher-order terms in the gravitational potential expansion for Jupiter when $r = 2R_e$.

($\theta = 90^\circ$) is apparent in the diagram. You should also note that these higher-order correction terms to a spherically symmetric potential are quite small; the first-order correction ($J_2 P_2$) is only on the order of a few tenths of a percent, the second-order term ($J_4 P_4$) is two orders of magnitude smaller than the first-order term, and the third-order term ($J_6 P_6$) is two orders of magnitude smaller than the second-order term.

Related to the gravitational moments is the moment of inertia of the planet. As was described in Problem 20.12, the moment of inertia is given by

$$I \equiv \int_{\text{vol}} a^2 dm, \quad (21.2)$$

where a is the distance of the mass element dm from the rotation axis (see Fig. 21.5). For an axially symmetric mass distribution, such as a giant planet rotating about a well-defined axis, it can be shown that I can be expressed in cylindrical coordinates as

$$I = 4\pi \int_{z=0}^{R_p} \int_{a=0}^{a_{\max}(z)} \rho(a, z) a^3 da dz, \quad (21.3)$$

where z is the distance from the center of the planet along the rotation axis to the point where a is measured out to dm , and R_p is the polar radius. If we assume that a cross section of the planet along the rotation axis can be approximated by an ellipsoid at the surface, then $a_{\max}$ is related to z by

$$\left(\frac{a_{\max}}{R_e}\right)^2 + \left(\frac{z}{R_p}\right)^2 = 1.$$

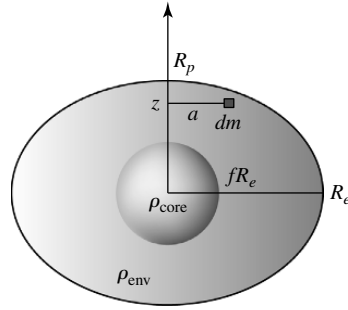


FIGURE 21.5 A model of an oblate planet with an elliptical cross section that has a spherical core. ρ_{env} is the density of the envelope and ρ_{core} is the density of the core. The transition between the two densities occurs at a fraction f of the equatorial radius of the planet.

For a two-component model of an oblate planet having an envelope density of ρ_{env} and a spherical core with a density of ρ_{core} , and where the transition between the two densities occurs at a fraction of the surface equatorial radius f , it can be shown that the moment of inertia is given by

$$I = \frac{8\pi}{15} R_e^4 [R_p \rho_{\text{env}} + f^5 R_e (\rho_{\text{core}} - \rho_{\text{env}})]. \quad (21.4)$$

Writing R_p in terms of the oblateness b of the planet, we have

$$R_p = R_e(1 - b),$$

Eq. (21.4) becomes

$$I = \frac{8\pi}{15} R_e^5 [(1 - b) \rho_{\text{env}} + f^5 (\rho_{\text{core}} - \rho_{\text{env}})]. \quad (21.5)$$

Clearly, the moment of inertia depends on the planet's oblateness and the mass distribution throughout the planet. Note that f cannot exceed $f_{\text{max}} = R_p/R_e = 1 - b \leq 1$.

The Cores of the Planets

All of these data suggest that Jupiter and Saturn have dense cores composed of a thick soup of “rock” (Mg, Si, Fe) and ices. However, although the data suggest dense cores, the masses of the cores are relatively poorly constrained. For example, note that Eq. (21.5) is strongly dependent on $f \leq 1$, and recall that the higher gravitational moments selectively sample the outer envelope of the planet. Based on the available data and numerical models, it appears that Jupiter probably has a rock/ice core of less than about $10 M_{\oplus}$, while Saturn's core may be about $15 M_{\oplus}$ with an uncertainty of perhaps 50%. (It is possible that the smaller core in Jupiter could be due to some portion of the core having eroded over the age of the planet.) Despite the core masses of Jupiter and Saturn being much greater than the mass of Earth, they constitute only a small fraction of the total mass of each planet. If we assume core masses of $10 M_{\oplus}$ and $15 M_{\oplus}$ for Jupiter and Saturn, respectively, their cores represent just

3% and 16% of the masses of the two gas giants. Hydrogen and helium make up most of the rest of the mass in each case.

Similar studies of Uranus and Neptune result in core masses comparable to those of Jupiter and Saturn: roughly $13 M_{\oplus}$ or so. However, in the cases of Uranus and Neptune, these cores constitute most of the mass of the planets. In particular, both Uranus and Neptune probably have 25% of their mass in the form of rock, 60% to 70% as “ices,” and only 5% to 15% in the form of hydrogen or helium gas. Clearly, Uranus and Neptune are not simply smallish versions of their larger siblings, and they are appropriately considered as ice giants rather than gas giants.⁵

Internal Heat and the Cooling Timescale

Another group of observations that provides hints about the formation and structure of the planets and their subsequent evolution is the detection of heat leaking out from the interior. In the terrestrial planets, the heat generated in the interior is due in large part to the slow decay of radioactive isotopes (page 753). However, this is not sufficient to account for the large quantities of heat coming from the interiors of the giant worlds. For example, as can be seen in Table 21.3, Jupiter absorbs (and re-emits) 5.014×10^{17} W of solar radiation, while 3.35×10^{17} W of additional power is produced in the interior of the planet. This significantly alters the energy balance and the thermal equilibrium temperature that would result from solar blackbody radiation alone. In Neptune’s case, more than one-half of the heat being radiated originates from the interior, explaining why its effective temperature is very close to that of Uranus, even though Neptune is much farther from the Sun.

One source of internal heat for the giant planets is the gravitational potential energy that was released by gases collapsing onto them during formation. This is just a consequence of the virial theorem (Section 2.4) and is the same Kelvin–Helmholtz mechanism that was discussed in Example 10.3.1 and Section 12.2.

Neglecting any slight differences due to composition and density, for a given specific heat capacity the total thermal energy content of a planet is proportional to its volume (i.e., $\propto R^3$). However, the rate at which heat leaves a planet by blackbody radiation is proportional to surface area ($\propto R^2$). Thus, without an additional source of energy, the timescale for cooling depends on radius as

$$\tau_{\text{cool}} = \frac{\text{total energy content}}{\text{energy loss/time}} \propto R^3/R^2 \propto R.$$

TABLE 21.3 Energy Budgets and Effective Temperatures of the Giant Planets. (Data from Table 2 of Guillot, *Annu. Rev. Earth Planet. Sci.*, 33, 493, 2005.)

Power or Temperature	Jupiter	Saturn	Uranus	Neptune
Absorbed power (10^{16} W)	50.14 ± 2.48	11.14 ± 0.50	0.526 ± 0.037	0.204 ± 0.019
Total emitted power (10^{16} W)	83.65 ± 0.84	19.77 ± 0.32	0.560 ± 0.011	0.534 ± 0.029
Intrinsic power emitted (10^{16} W)	33.5 ± 2.6	8.63 ± 0.60	0.034 ± 0.038	0.330 ± 0.035
Effective temperature (K)	124.4 ± 0.3	95.0 ± 0.4	59.1 ± 0.3	59.3 ± 0.8

⁵The term *ices* in this context is somewhat misleading since the H_2O , CH_4 , NH_3 , and other constituents are actually in a somewhat fluid state under the high pressures found in the interiors of the giant planets.

The characteristic time required for a planet to cool is roughly proportional to the planet's radius. Extrapolating back in time, the giant planets must have been much more luminous when the Solar System was in its infancy; Jupiter may have even glowed visibly.

Since Jupiter is larger than Saturn (as well as being closer to the Sun), it should have remained hotter for a longer period of time and should still be radiating energy into space at a greater rate. In Saturn's case, however, the energy available from the primordial collapse is not sufficient to account for all of the heat now observed to be coming from the planet. The solution to the puzzle of Saturn's additional heat source lies in the observation that its helium is significantly depleted in the upper atmosphere. Referring to Table 21.1, note that helium only accounts for about 3% of the particles in Saturn's upper atmosphere, while the value is closer to 16% for Jupiter and nearly 20% for the Sun. The slow sinking of the heavier helium atoms relative to hydrogen through the atmosphere causes a change in the gravitational potential energy of the planet and the accompanying generation of heat via the virial theorem. This effect has been more pronounced in Saturn because the planet is somewhat cooler.

Modeling the Interiors of the Giant Planets

Modeling the interiors of the giant planets is done in much the same way it is done for stars; the major difference is the kind of material used in their construction. For example, at the relatively cool temperatures and high pressures of the giant planet interiors, hydrogen takes on a very strange form by terrestrial standards. As we move deeper into the planet, the familiar form of molecular hydrogen becomes so compressed that the molecular bonds are broken and the orbital electrons become shared among the atoms. This is very similar to the behavior of a metal; the hydrogen inside the planet takes on the characteristics of a molten metal, much like mercury at room temperature. This exotic equation of state of hydrogen has been verified in terrestrial laboratories by creating shock waves in the gas that produce temperatures of several thousand kelvins and pressures of millions of atmospheres. It appears that **liquid metallic hydrogen** actually dominates the interiors of Jupiter and Saturn. For Uranus and Neptune, the pressures probably do not get large enough to convert hydrogen into its liquid metallic form, but the ices present in their atmospheres (such as methane and ammonia) become ionized by the pressure.

The interior structures of the giant planets are depicted in Fig. 21.6. The regions labeled "inhomogeneous" for the gas giants are where helium becomes insoluble in hydrogen, and helium-rich droplets form. These droplets then sink deeper into the planet, releasing gravitational potential energy. In the case of Saturn the helium may have settled into the core or formed a shell around the core. Uranus and Neptune have very little hydrogen and helium and are dominated by ices and rock.

The Upper Atmospheres

In their upper atmospheres, the very colorful and dynamic cloud tops of Jupiter, the more muted hues of Saturn, and the deep blue-greens of Uranus and Neptune owe their beauty to the temperature, composition, rotation, and internal structures of the planets. Observational data, combined with theoretical modeling, suggest that Jupiter's clouds exist in three layers. Clouds in the top layer are composed of ammonia, the next layer is probably composed of ammonium hydrosulfide, and the clouds in the deepest layer are made of water.

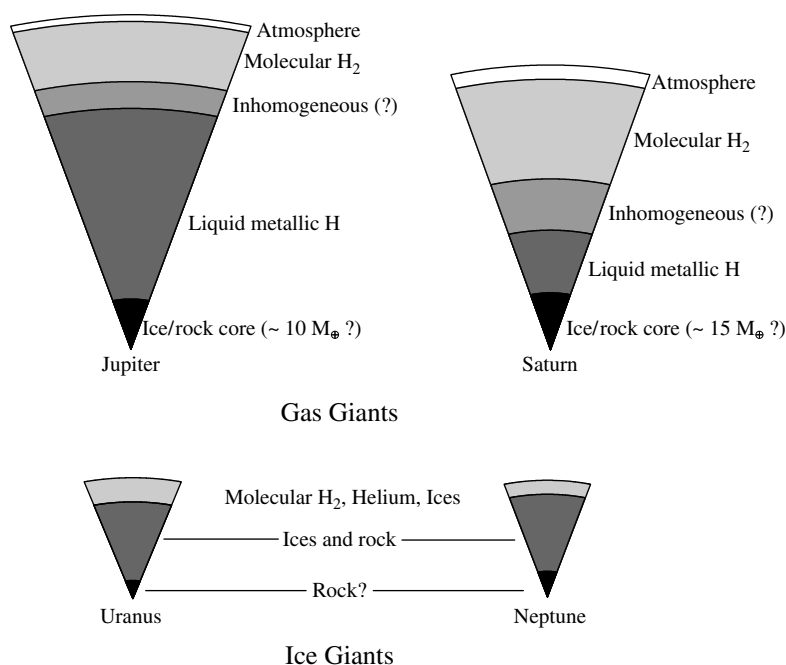


FIGURE 21.6 Computer models of the interiors of the giant planets. The relative sizes of the planets are depicted correctly. (Adapted from Guillot, *Annu. Rev. Earth Planet. Sci.*, 33, 493, 2005.)

The coloration in the clouds of both Jupiter and Saturn is caused by their atmospheres' compositions, although which colors are associated with which molecules remains unclear; suggestions include sulfur, phosphorus, or various organic (carbon-rich) compounds. In Jupiter and Saturn, the bluish regions apparently have higher temperatures, indicating that they lie deeper in the atmosphere. At progressively higher altitudes are brown, white, and red clouds.

Overall, the clouds are located deeper in the atmosphere of Saturn when compared with Jupiter, and hence are not as dramatic. In Uranus and Neptune, reflective clouds of ammonia and sulfur are located deep in the atmosphere. As sunlight passes through the atmosphere, the blue wavelengths are scattered most efficiently by the molecules. In addition, the presence of methane in the atmosphere tends to absorb the red light.

The Comet P/Shoemaker–Levy 9 Impacts on Jupiter

During July 16–22, 1994, Jupiter took center stage as it got pummeled by the fragments of **Comet P/Shoemaker–Levy 9 (SL9)**.⁶ The comet was discovered in March 1993, although it had apparently been orbiting Jupiter for decades. Extrapolating the comet's orbit back in time, it appears that SL9 broke apart on July 8, 1992, as it passed within 1.6 R_J of Jupiter, well within the planet's Roche limit. [A Hubble Space Telescope view of 21 fragments is

⁶The nature of comets will be discussed in some detail in Section 22.2.

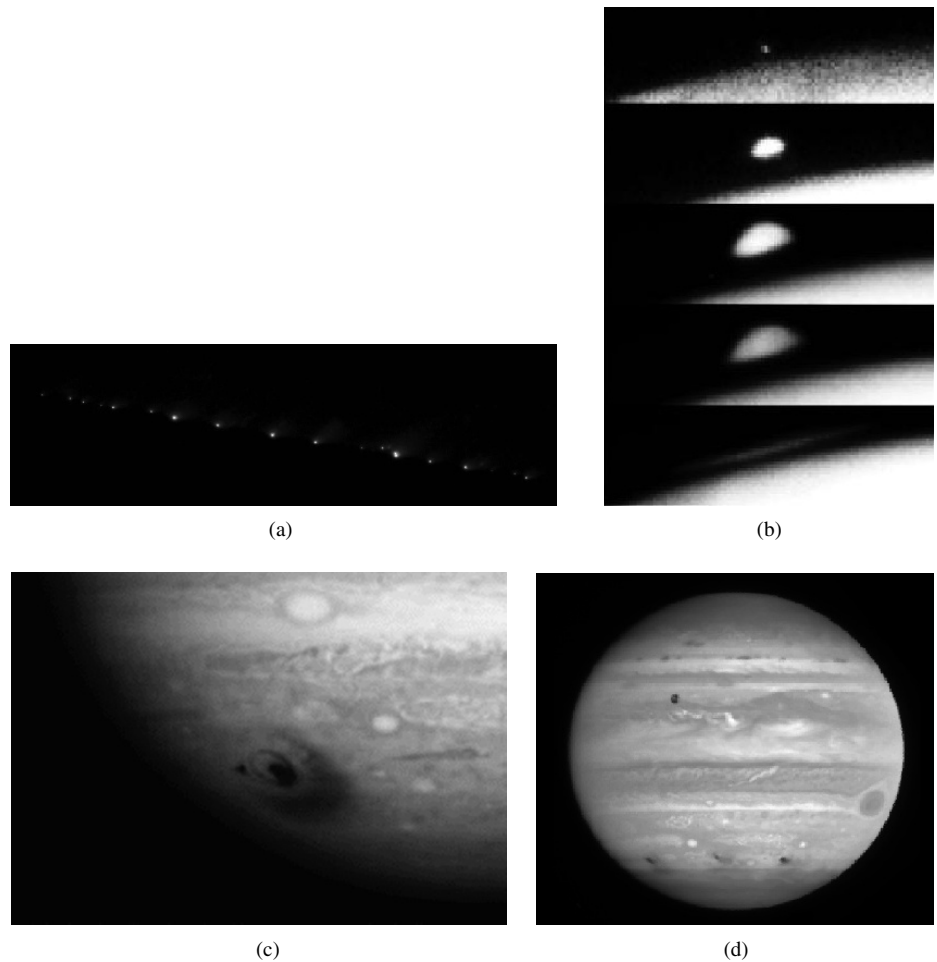


FIGURE 21.7 (a) 21 fragments of SL9 seen on May 17, 1994. The line of cometary nuclei stretches for 1.1×10^6 km. [Courtesy of H. A. Weaver, T. E. Smith (Space Telescope Science Institute), and NASA.] (b) Hubble Space Telescope images, taken several minutes apart, showing the plume from fragment G on July 18, 1994. (c) Close-up of the fragment G impact site. (Courtesy of Dr. Heidi Hammel, Massachusetts Institute of Technology, and NASA HST.) (d) From left to right, the impact sites in the southern hemisphere of fragments C, A, and E. One of Jupiter's moons (Io) can be seen crossing the planet's disk. (Courtesy of the Hubble Space Telescope Jupiter Imaging Team.)

shown in Fig. 21.7(a).] Astronomers soon realized that the comet fragments would crash into Jupiter in July 1994, possibly providing important clues to the nature of comets and the structure of Jupiter's atmosphere.

Over the week when the collisions occurred, virtually all of the telescopes on Earth (including amateur telescopes) that were in position to view the event, as well as space-based observatories such as the Hubble Space Telescope, Galileo, and Voyager 2, were focused on

Jupiter. Various predictions had indicated that some direct evidence of the impacts might be observable from Earth, but the spectacular display that ensued far exceeded expectations. Figure 21.7(b) shows several images of the enormous plume that rose 3500 km above the cloud tops when fragment G (believed to be the largest) entered the atmosphere of the planet. Even though each of the collisions occurred just beyond our view, on the side of Jupiter away from Earth, the plumes were high enough to make them visible above the limb.⁷ The fireballs reached temperatures of 7500 K, greater than the effective temperature of the Sun. Data for the G impact indicate that the temperature cooled to 4000 K after five seconds. Analysis of the data indicated that the largest fragments were no more than 700 m across.

Immediately after each of the larger collisions, scars appeared in the atmosphere greater in diameter than Earth [see Figs. 21.7(c) and (d)]. The dark nature of the marks was probably due to organic molecules rich in sulfur and nitrogen that were present in the atmosphere before the collision. It is also possible that some of the coloration was due to carbon-based compounds, like graphite, that contained silicates delivered by the comet fragments. By December 1994, the marks had been torn apart by the motions in Jupiter's atmosphere, forming a ring around the planet that eventually dissipated completely.

Atmospheric Dynamics

The most famous atmospheric feature on Jupiter is its **Great Red Spot**, apparent in Figs. 21.1(a) and 21.8(a). This huge anticyclonic storm, which measures roughly one Earth diameter wide by two Earth diameters long, has been observed for more than three centuries. Smaller but similar features can be seen in the atmospheres of each of the giant worlds. Another characteristic shared by these planets is the banded cloud structure following lines of constant latitude. In the case of Uranus the banded cloud features are very difficult to

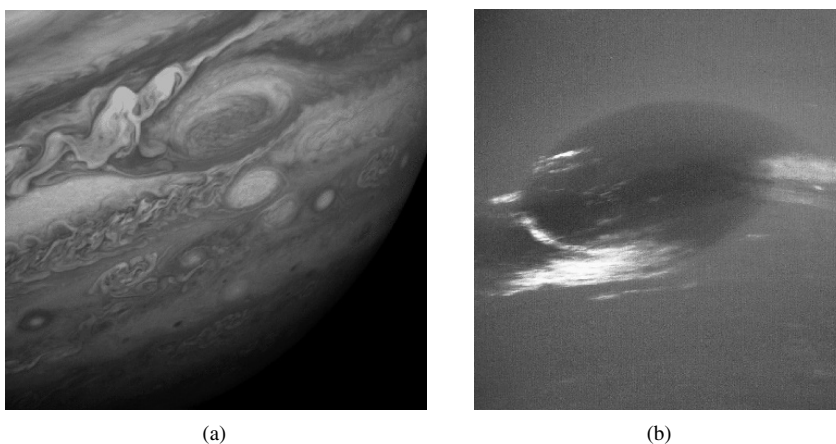


FIGURE 21.8 (a) The Great Red Spot of Jupiter. (b) The Great Dark Spot of Neptune. (Courtesy of NASA/JPL.)

⁷Only the Galileo and Voyager 2 spacecraft had direct views of the impacts.

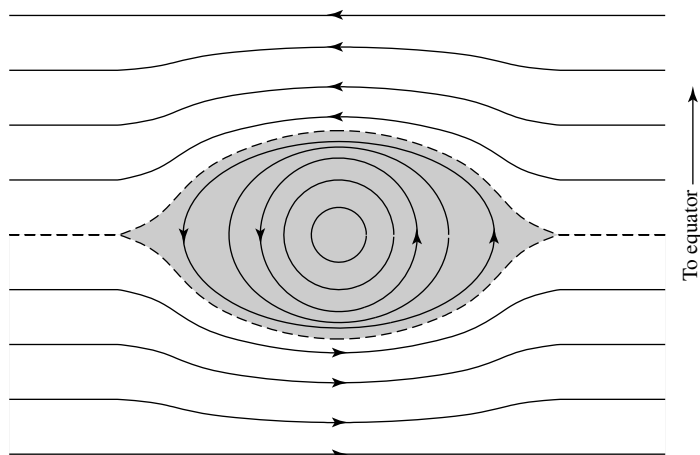


FIGURE 21.9 Circulation around Jupiter's Great Red Spot is counterclockwise. The anticyclonic storm (which is in the southern hemisphere) is located between two atmospheric bands that are moving in opposite directions. Wind speeds within the Great Red Spot reach 100 m s^{-1} , and eddies at the edge of the system circle it in about 7 days.

detect, but they do exist. The circulation within Jupiter's Great Red Spot is attributable to its location between two rivers of atmosphere moving in opposite directions (see Fig. 21.9).

Despite these apparently long-lived features, the atmospheres are very dynamic, with rapid changes occurring on small scales, including rotation around the more stable cyclonic structures. It is worth noting, however, that large features are not necessarily permanent either. For instance, when Voyager 2 visited Neptune in 1989, it discovered the **Great Dark Spot** in the southern hemisphere, shown in Figs. 21.1(d) and 21.8(b). Later, when the planet was observed again by the Hubble Space Telescope in 1994, the Great Dark Spot was gone. Then, in 1995, another dark spot appeared in the northern hemisphere.

Just as the Coriolis force (page 730) redirects the large-scale circulation in Earth's atmosphere from north–south to predominantly east–west flow patterns in each hemisphere, Hadley circulation in the more rapidly rotating giant planets (particularly Jupiter and Saturn) is similarly redirected. However, the atmospheric circulation of Uranus has an interesting aspect not shared with the other giant planets. Unlike any other planet in the Solar System except Pluto, Uranus is almost lying on its side; its rotation axis is tilted 97.9° to the ecliptic. This implies that each pole has the Sun overhead for a portion of its 84-year orbit. During those periods one would expect heat to be transported from the subsolar pole to the one that is in darkness. Yet, when Voyager 2 passed Uranus in 1986, at about the time when one of the poles was pointed toward the Sun, the visible flow patterns were still largely parallel to the planet's equator, due to the planet's rapid rotation and the effects of the Coriolis force. How Uranus was able to transport heat away from the subsolar pole without detectable pole-to-pole flow patterns remains an open question.

Another noticeable difference between Uranus and the rest of the giant worlds is its lack of prominent vortices. This may correspond to the lack of any detectable heat flow outward

from the deep interior. Although it must certainly exist, the rate of heat flow is clearly much less pronounced than in the other three giants.

Magnetic Fields

The molten iron–nickel core of Earth is the source of its magnetic field. In the giant planets, it is liquid metallic hydrogen that appears to fill that role, at least in Jupiter and Saturn. Rapid rotation generates electric currents in the conducting interiors of the planets. Because the magnetic fields are almost certainly anchored deep in their interiors, measuring the rotation periods of the fields provides a method of determining the rotation periods of their interiors.

In the 1950s, measurements of the radio-wavelength radiation being emitted from Jupiter revealed both thermal and nonthermal components. The thermal radiation is just part of the energy being given off by the planet itself (blackbody radiation). However, the strong nonthermal component was determined to be synchrotron radiation (see Section 4.3) with wavelengths in the decameter (tens of meters) and the decimeter (tenths of meters) ranges. This implies that Jupiter must have a significant magnetic field with relativistic electrons trapped in it. The measured strength of the field is some 19,000 times greater than Earth's field.

Another interesting consequence of the SL9 collisions in Jupiter's southern hemisphere [which all occurred at nearly the same latitude; see Fig. 21.7(d)] was the appearance of an auroral display in the northern hemisphere, not unlike the aurorae seen on Earth; recall Fig. 20.15. Apparently, charged particles near the collision sites acquired sufficient kinetic energy that they traveled along Jupiter's magnetic field lines, colliding with the atmosphere in the north within 45 minutes following the impacts.

The physical extent of Jupiter's magnetic field is enormous. The planet's **magnetosphere**, defined to be the space enveloped by its magnetic field, has a diameter of 3×10^{10} m, 210 times the size of the planet and 22 times larger than the Sun. Because of Jupiter's rapid rotation, the charged particles trapped in its field are spread out into a **current sheet** that is situated along *the field's* equator (the field axis is inclined 9.5° to the rotation axis of the planet). Given the large numbers of particles present in Jupiter's current sheet, another source of charged particles beyond those supplied by the solar wind must exist. The solution to this mystery came when the Voyager spacecraft first observed Jupiter's moon Io.

21.2 ■ THE MOONS OF THE GIANTS

Many of the most spectacular and fascinating images returned by the Voyager, Galileo, and Cassini–Huygens missions were of the moons of the giant planets, beginning with the Galilean moons of Jupiter (Fig. 21.10). The relative sizes of the Galilean moons are depicted in Fig. 21.11. **Io** (shown in more detail in Fig. 21.12) is the closest of the four large Galilean moons to Jupiter. It is a bizarre-looking yellowish-orange world with as many as nine active volcanoes observed to be erupting simultaneously. **Europa** (Fig. 21.13) is covered with a thin layer of water-ice that is criss-crossed by cracks and nearly devoid of any cratering. **Ganymede** (Fig. 21.14) has a thick ice surface that shows evidence of significant cratering. And finally, **Callisto** (Fig. 21.15) appears to be covered with a layer of dust and has an old and



FIGURE 21.10 A “family portrait” of Jupiter and its four largest moons. From nearest Jupiter to farthest are Io, Europa, Ganymede, and Callisto. The portrait is actually a mosaic of a number of Voyager images. (Courtesy of NASA/JPL.)



FIGURE 21.11 A mosaic of images obtained by the Galileo spacecraft showing the four Galilean moons of Jupiter. From left to right, and from nearest to Jupiter to farthest away: Io, Europa, Ganymede, Callisto. Here, the moons are depicted in such a way as to show their relative sizes. (Courtesy of NASA/JPL.)

very thick ice crust that has been subjected to extensive bombardment.⁸ The characteristics of these worlds are consistent with a decreasing average density with increasing distance from Jupiter, implying that the relative amount of water-ice crust increases with respect to the rock core.

The Evolution of the Galilean Moons

The increasing percentage of volatiles (principally water-ice) in these worlds at increasing distances from Jupiter suggests that their formation was closely linked to the formation and

⁸Recall the discussion in Section 20.4 describing the amount of cratering as a function of surface age.

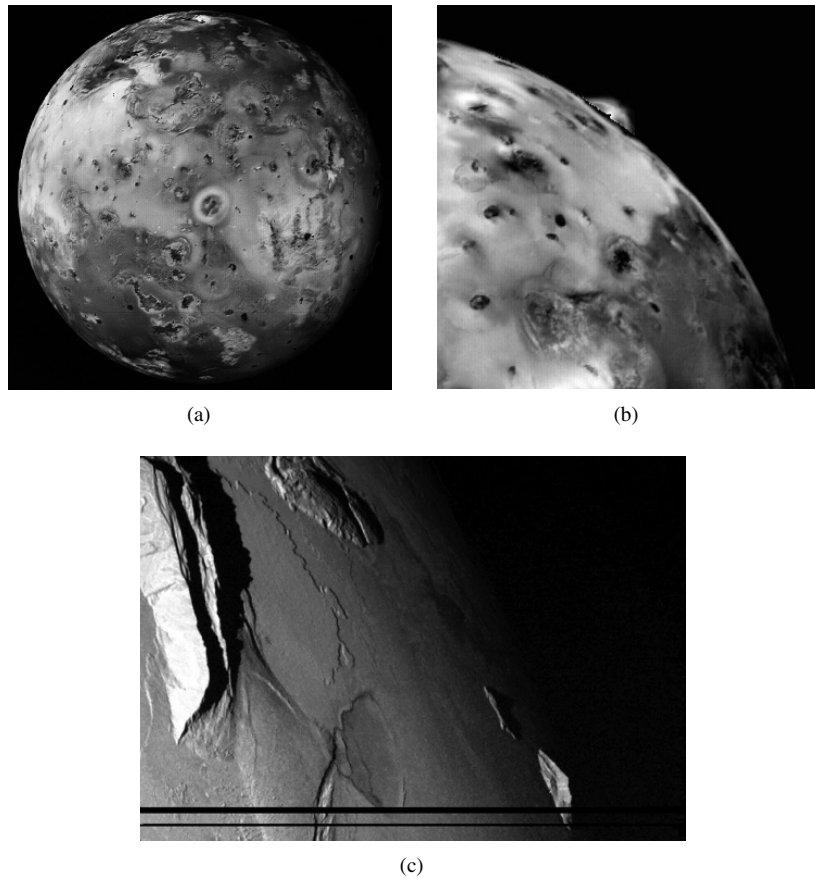


FIGURE 21.12 (a) The disk of Io shows a large number of volcanic features. (b) A volcano (Prometheus) erupting on the limb of Io. Prometheus was observed to be erupting in every image obtained by the Voyagers (1979) and Galileo (1995–2003). Other volcanic eruptions are not as long-lived. (c) Mountains on Io seen at sunset. The low scarp in the upper left is approximately 250 m high. It is believed that these mountains are produced by uplifted thrust faults. The black lines along the bottom of the image are due to missing data. (Courtesy of NASA/JPL/University of Arizona/Arizona State University.)

subsequent evolution of the planet itself. Given the regular nature of the Galilean satellites, it has been proposed they may have formed out of the Jupiter *subnebula* while Jupiter was accreting its massive atmosphere. Within this context, recalling that Jupiter must have been hotter in the past than it is today, Io would have been close enough to have had most of its volatiles evaporate away. Moving progressively farther out, Europa would have been able to hold on to some water, Ganymede even more, and Callisto (being the coldest of the Galilean moons at the time of its formation) would have retained the largest percentage of volatiles.

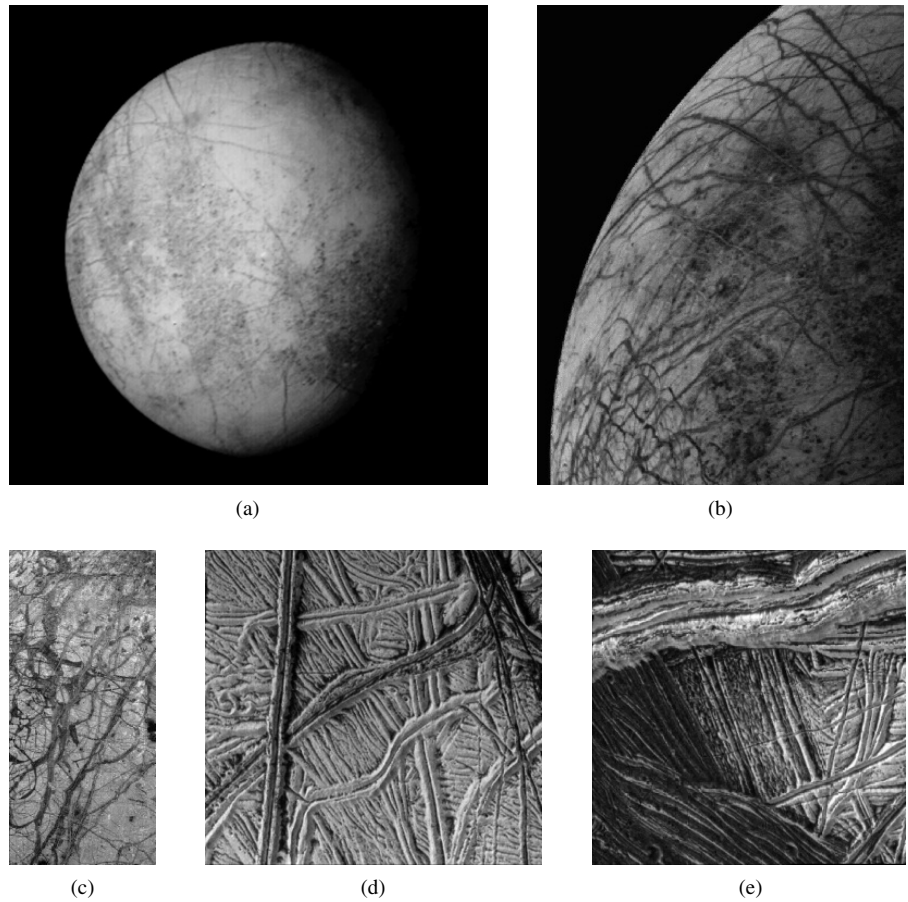


FIGURE 21.13 (a) The full disk of Europa. (b) Europa has numerous cracks running across the surface. (c) A close-up of broken ice. (d) Ridged plains. (e) Wedge terrain. (Courtesy of NASA/JPL.)

The Effects of Tidal Forces on Io

The consequences of this evolution can be seen in each of the Galilean moons. Consider them in sequence beginning with the one closest to Jupiter. Because of its proximity to Jupiter, Io experiences the most severe tidal forces. Even though the moon's rotation period is the same as its orbital period, small deviations from a perfectly circular orbit mean that its orbital velocity is not constant. Consequently, the moon tends to wobble, not quite keeping one side "locked in place" toward Jupiter. This effect is due to the curious resonance that exists among the orbits of Io, Europa, and Ganymede. Their orbital periods form ratios that are approximately 1:2:4, meaning that both Europa and Ganymede perturb Io's orbit at about the same location each time Io orbits the planet. This forces Io's orbit to remain slightly elliptical.

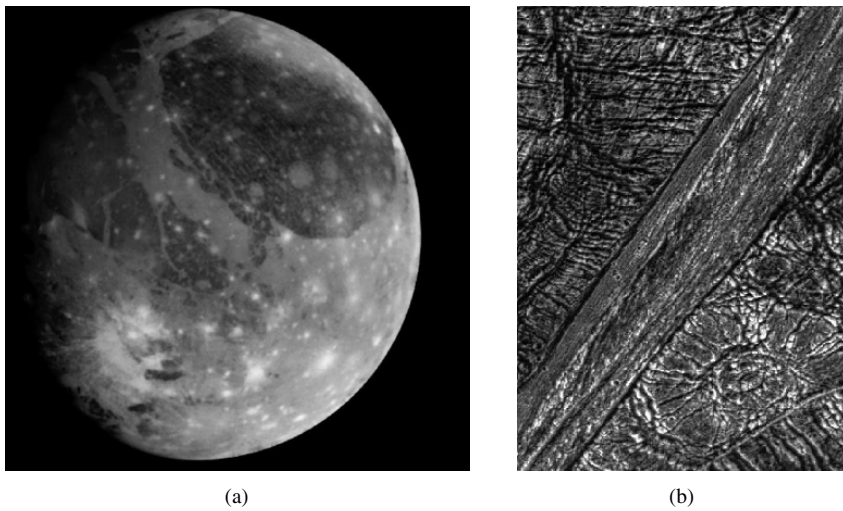


FIGURE 21.14 (a) The surface of Ganymede shows significant cratering, indicating that it has not been refreshed as recently as Europa. (b) A close-up view of ridges and grooves prevalent on the surface, indicative of past tectonic activity. The diagonal band is 15 km wide. The circular feature in the lower right portion of the image is probably an impact crater. (Courtesy of NASA/JPL.)

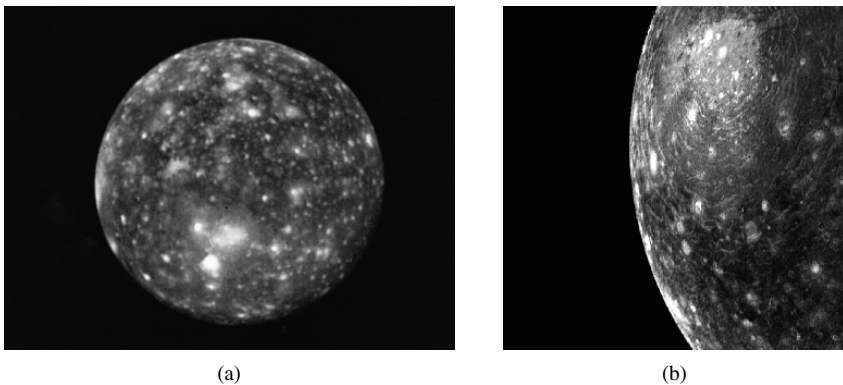


FIGURE 21.15 (a) Callisto's surface exhibits extensive cratering. (b) A close-up of a large impact crater known as Valhalla. (Courtesy of NASA/JPL.)

Based on gravitational data from Galileo's close flybys of Io, it appears that Io has an iron-rich core, a molten silicate mantle, and a thin silicate crust (the moon's average density is 3530 kg m^{-3}). This structure suggests that Io was entirely molten at least once, and perhaps numerous times, allowing the moon to become chemically differentiated. Lava flows and lakes of lava, such as Loki Patera (which is larger than the island of Hawaii) are clearly evident on the surface of Io. It is important to note, however, that Io's volcanoes do not operate in quite the same way that Earth-based volcanoes do. Instead, the eruptions may be

somewhat more analogous to the geysers seen in such places as Yellowstone National Park. In terrestrial geysers, rapid phase transitions from water to steam force the steam upward at high velocity through cracks in the surface. On Io, sulfur and sulfur dioxide (SO_2) probably play the same role. In fact, SO_2 has been detected over volcanic vents and in Io's very thin atmosphere. The yellowish-orange surface is due to sulfur raining back down on the moon from its continually erupting volcanoes. The constant eruptions are always regenerating the moon's surface; Io is literally turning itself inside out.

Io's Interaction with Jupiter's Magnetic Field

All of the Galilean moons are located deep inside Jupiter's magnetosphere, but Io interacts with the magnetic field most strongly. Since Jupiter rotates in just under 10 hours, whereas Io orbits the planet in 1.77 days, Jupiter's magnetic field sweeps past Io at a speed of about 57 km s^{-1} . This motion through the magnetic field sets up an electric potential difference across the moon, estimated to be 600 kV.⁹ The potential difference acts much like a battery, causing a current of nearly 10^6 amps to flow back and forth along magnetic field lines between Io and Jupiter. This current flow of charged particles in the magnetic field also generates Joule heating within the moon, analogous to a resistor in a circuit. Roughly $P = IV \sim 6 \times 10^{11} \text{ W}$ is generated in this way. However, this contribution to the total internal heating of the moon is only a small fraction of the total energy liberated from the surface per second, which is approximately 10^{14} W .

That Io must have some interaction with Jupiter's magnetic field has been known for some time. When Jupiter, Io, and Earth are in certain alignments, bursts of decameter-wavelength radiation are detected. Not all the details of the process are yet understood, but the bursts appear to be associated with the electrical current flowing between Jupiter and its volcanic moon.

Io must also be responsible for the excessive number of charged particles trapped in Jupiter's magnetic field, although it is unlikely that they escaped directly from the moon's volcanoes since the ejection speeds are much less than Io's escape velocity. Instead, a process referred to as *sputtering* has been proposed; oxygen and sulfur ions from Jupiter's magnetosphere impacting on the moon's surface or in its atmosphere may provide sufficient energy for other sulfur, oxygen, sodium, and potassium atoms to escape. In fact, clouds of sulfur and sodium (known as the **Io torus**) have been detected around Jupiter at the location of Io's orbit. On the order of 10^{27} to 10^{29} ions leave Io and enter Jupiter's magnetospheric plasma every second.

Europa

Europa's surface seems to be continually refreshed. Based on the near absence of cratering, it appears that most of the surface is less than 100 million years old, supporting the idea that a layer of liquid water may exist below the surface. In fact, observations from the Galileo mission indicate an iron-rich core, a silicate mantle, a possible subsurface ocean, and a thin ice crust; the average density of Europa is 3010 kg m^{-3} , less than the density of Io. The water ocean/ice crust is collectively about 150 km thick. The source of the heat required to

⁹This is just Faraday's law of induction.

keep the subsurface water at least partially melted is probably the weak tidal interactions with Jupiter and the other Galilean moons. The cracks running across the moon's surface appear to be stress fractures induced by those tidal forces combined with tectonic activity.

In 1994 the Hubble Space Telescope detected a thin molecular oxygen atmosphere around Europa. That observation was confirmed by both Galileo and the flyby of the Cassini spacecraft on its way to a rendezvous with Saturn. Cassini also found the presence of atomic hydrogen in Europa's atmosphere. It has been suggested that the atmosphere is due to sputtering of the surface water-ice resulting from the interaction of Europa with Jupiter's magnetosphere.

Given the presence of subsurface heating, a likely source of liquid water, and the probability of organic materials being either intrinsic to the moon or delivered by comets and meteorites, it has been widely speculated that Europa may be a site for the evolution of life. Even though no evidence exists for the presence of life today or in the past on Europa, at the end of its operational lifetime scientists intentionally sent the Galileo spacecraft into the crushing atmosphere of Jupiter on September 21, 2003, in order to avoid any future inadvertent collision with Europa and its possible subsurface ocean.

Ganymede

Ganymede's surface also shows a complex series of ridges and grooves that strongly suggest some history of tectonic activity on this ice world. This is supported by gravitational data from Galileo indicating a likely partially molten iron core, a silicate lower mantle, an icy upper mantle, an ice crust, and an average density of only 1940 kg m^{-3} . It has been proposed that before the ice crust became too rigid, convection in the interior was responsible for carrying heat to the surface. This convective motion also caused movement of the surface crust, much like the current action of tectonic plates on Earth. As a result, although the surface is certainly much older and more heavily cratered than Europa's, the surface has been at least partially refreshed during its history.

Callisto

Callisto apparently cooled and solidified quite rapidly after material accreted out of the local subnebula around Jupiter. As a result, its surface continued to collect dust as the nebula thinned, blanketing the moon with the dark material. Evidence that Callisto solidified quickly is also apparent in the structure of its interior. Models suggest that the interior of the moon is relatively simple, with a partially differentiated interior of ice and rock, an ice-rich crust, and the lowest density of the Galilean moons (1830 kg m^{-3}). Having solidified in the early stages of the formation of the Solar System, Callisto was also subject to frequent impacts of the still-abundant objects that traveled among the newly formed planets and moons. Evidence of the nebular dust accretion and the impacts remains today. The whitish-appearing impact craters are the result of ice being exposed during the collisions.

A Unified Formation of the Galilean Moons

As we have seen, the four Galilean satellites of Jupiter exhibit a trend of decreasing density with distance from their parent planet. Given the evident trends in their properties, including their internal structure (diminishing iron-rich cores and increasing water-ice content with

distance from Jupiter), it is apparent that they likely formed systematically with Jupiter, perhaps out of a subnebula around the planet. This is also supported by the fact that each of the Galilean moons orbit prograde and in the equatorial plane of the planet.

The Smaller Moons of Jupiter

Other, smaller moons also orbit in a prograde direction in the equatorial plane of Jupiter. These additional **regular satellites** may have formed out of the subnebula as well. However, there are a large number of moons around Jupiter that orbit well out of the equatorial plane and, in many cases, in retrograde orbits. It seems that these **irregular satellites** of Jupiter may be captured objects that happened to wander by at some point in time. Still other very small satellites may be **collisional shards** produced by meteoritic collisions with larger satellites.

Space does not allow us to discuss each of these many moons in detail. Similarly, we are unable to discuss most of the smaller moons orbiting the other giant planets; rather, we will focus our attention on the larger moons and a few of the more unusual smaller satellites of Saturn, Uranus, and Neptune.

Saturn's Titan with Its Thick Atmosphere

When the two Voyager spacecraft reached Saturn in 1980 and 1981, they were directed to examine **Titan**, the second-largest moon in the Solar System (Ganymede being the largest). Ever since Gerard P. Kuiper (1905–1973) detected methane gas around Titan in the 1940s, astronomers have wondered about the nature of this distant, atmospheric world. When the images began arriving, scientists saw a moon with an atmosphere so filled with suspended particles (**aerosols**) that no pictures of its obscured surface were possible.

The joint Cassini–Huygens mission arrived in the Saturnian system in July 2004. After arrival, the Huygens probe detached from the Cassini orbiter and descended to the surface of Titan on January 14, 2005. During its descent, the Huygens probe was able to measure wind speeds of up to 210 m s^{-1} , sample the composition of the atmosphere, and, after passing through the high-altitude smog layer of hydrocarbons, obtain images of the surface (see Fig. 21.16).

The dominant constituent in the atmosphere is nitrogen (N_2), which constitutes somewhere between 87% and 99% of the gases. Methane (CH_4) makes up between 1% and 6% of the atmosphere, and argon (Ar) constitutes between 0% and 6% of the total. Numerous other species are present in smaller amounts, including molecular hydrogen (H_2), carbon monoxide (CO), carbon dioxide (CO_2), hydrogen cyanide (HCN), and a host of additional hydrocarbons, such as acetylene (C_2H_2), ethylene (C_2H_4), ethane (C_2H_6), methylacetylene (C_3H_4), propane (C_3H_8), and diacetylene (C_4H_2). The aerosols in the high-altitude smog layer are probably just condensed forms of these compounds.

At the base of the atmosphere, the pressure is approximately 1.5 atm and the temperature is 93 K. With those conditions, methane is able to condense as a liquid and then evaporate again, and thus it plays a role much like that of water on Earth. At the Huygens landing site, the ground was moist, with liquid methane occurring a few centimeters below the surface. It is possible that it had rained methane at that location shortly before the arrival of Huygens. In fact, Huygens sank 10 to 15 cm into the soft ground at the landing site. The

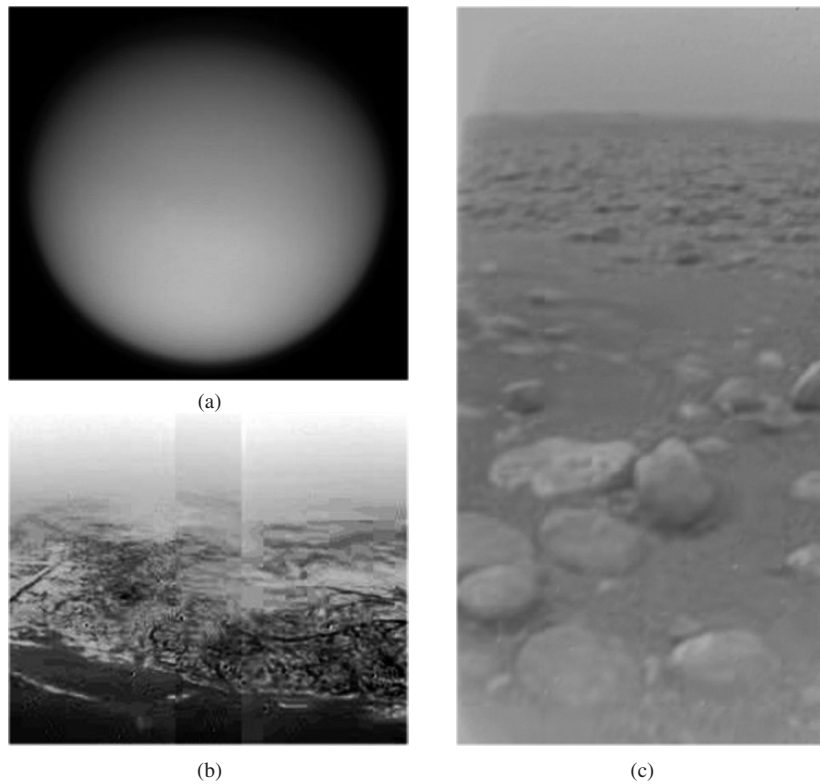


FIGURE 21.16 Counterclockwise from upper left: (a) Titan with its thick atmosphere as seen by the Cassini orbiter (Courtesy of NASA/JPL). (b) A mosaic image of the surface from an altitude of 8 km imaged by the Huygens probe (Courtesy of ESA/NASA/JPL/University of Arizona). (c) The surface of Titan with “pebbles” of what is probably water-ice in the foreground. The flat pebble near the middle of the image has a width of 15 cm, and the one to its right has a width of 4 cm. Both pebbles are 85 cm from the Huygens probe camera. (Courtesy of ESA/NASA/JPL/University of Arizona.)

surface water-ice pebbles also show evidence of having a liquid flow across them, much like terrestrial rock pebbles in a dry creek bed. In addition, images obtained during the descent revealed topography that looks like drainage canals leading to low, dark, flat regions that may be lakes (or dried lake beds).

Mimas and the Herschel Crater

Another member of the Saturnian system, **Mimas**, is a small but fascinating moon (shown in Fig. 21.17). It exhibits a very large impact crater (referred to as Herschel) that is testimony to a collision almost energetic enough to fracture it.¹⁰ Of course Saturn also has numerous

¹⁰More than one researcher has noticed that Mimas bears a strong resemblance to the “Death Star” in the George Lucas film *Star Wars*, Lucasfilm Ltd. Production (1977).



FIGURE 21.17 The impact that produced the Herschel crater on Mimas was nearly energetic enough to completely fracture the moon. Mimas is one of the many small moons orbiting Saturn. (Courtesy of NASA/JPL.)

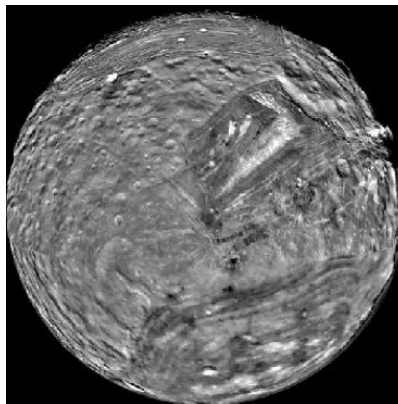


FIGURE 21.18 Miranda is one of the moons of Uranus. Its dramatic surface features may be the result of one or more very energetic collisions that fractured the moon. (Courtesy of NASA/JPL.)

other regular and irregular satellites in its system, a few of which will be mentioned in the context of its extensive ring system in Section 21.3.

The Chaotic Surface of Miranda, a Moon of Uranus

When Voyager 2 reached Uranus in 1986, it encountered another moon that may have suffered a very energetic collision. **Miranda**, which measures only 470 km across, looks like a moon “put together by a committee” (Fig. 21.18). One explanation for its amazing topography is that one or more collisions actually succeeded in breaking the moon apart. When gravity pulled all of the pieces back together, they didn’t quite fit. Portions of the rock core tried to settle back to the center of the moon while ice tried to float back to the surface. This proposed rearrangement of the structure of Miranda produced a strange surface with

cliffs as tall as 20 km (twice the height of Mount Everest) and features such as the “chevron” that can be seen in the figure.

An alternative explanation for Miranda’s topography proposes that tidal forces exerted by Uranus on the small moon caused parts of the surface to be pulled apart. This allowed warmer material in the interior (which was heated by tidal effects) to rise to the surface, producing the ridges and troughs that are observed.

Interestingly, all of the regular moons of Uranus, and, as we will see in the next section, its ring system as well, orbit near the equatorial plane of the planet rather than its orbital plane. Recall that the rotation axis of Uranus is highly inclined to the ecliptic (97.9°), making the orientation of the Uranian system a puzzle for Solar System dynamicists.

Neptune’s Triton

The last and one of the most unusual moons visited by Voyager 2 was Neptune’s largest moon, **Triton** (Fig. 21.19). With a surface temperature of 37 K, it is also the coldest world yet visited. The moon’s southern pole is covered with a pinkish frost that is composed almost entirely of nitrogen. Along with the nitrogen frost, other surface ices include CH_4 , CO , and CO_2 . Also present are very large “frozen lakes” of water-ice that show very little cratering, indicative of a relatively young age. The water-ice may have erupted from ice volcanos.

During the Voyager 2 flyby, geyser-like jets were detected forcing plumes of gas 8 km up into Triton’s tenuous atmosphere, where the plumes were blown down-wind. These plumes may simply be gas rising from a warm source inside the planet, but how they are initiated remains unclear.

The atmosphere of Triton is composed predominantly of nitrogen, like the atmospheres of Earth, Titan, and Pluto. However, unlike Earth and Titan, Triton’s atmosphere is extremely thin, with a pressure of only 1.6×10^{-5} atm. Much of the atmosphere may be a consequence of the jets of nitrogen gas erupting from the interior of the moon.

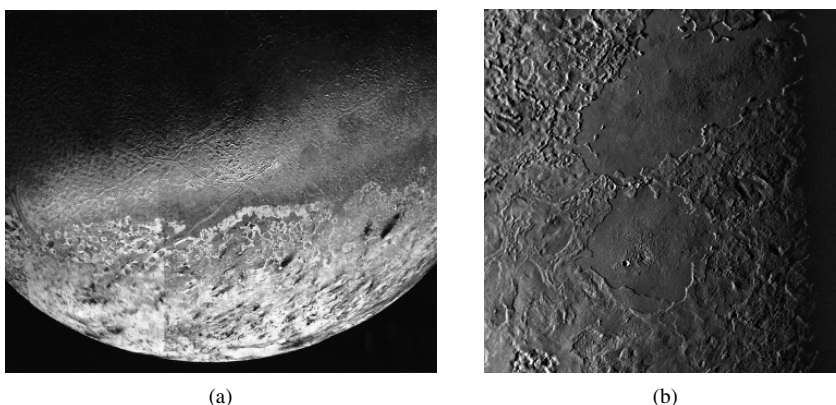


FIGURE 21.19 (a) The southern polar ice cap of Triton, Neptune’s largest moon. The dark streaks appear to be a mixture of nitrogen frost and hydrocarbons that was ejected from small volcanos. (b) A water-ice lake that may have been produced by an ice volcano. Given the lack of significant cratering, the surface appears to have been refreshed relatively recently. (Courtesy of NASA/JPL.)

Triton's slowly decaying retrograde orbit was discussed on page 723. Given its highly unusual orbit, both retrograde and inclined to Neptune's equator by 20° , its proximity to the Kuiper Belt, and physical properties similar to other Kuiper Belt objects (such as Pluto), it is widely believed that Triton was captured by Neptune. It is also likely that when the relatively massive Triton was captured, it significantly disrupted Neptune's already-existing system of much smaller satellites. Perhaps the tidal effects that resulted in the moon's present-day circularized and synchronous orbit could have produced sufficient internal heating to cause faulting and its cantaloupe-like terrain.

21.3 ■ PLANETARY RING SYSTEMS

Each of the giant planets contains a ring system. Although Saturn's prominent rings were first seen several hundred years ago, the rings of the other planets weren't discovered until the 1970s and 1980s. As we will learn, there are certainly some similarities among the ring systems, but there are significant differences as well.

The Structure of Saturn's Rings

Arguably, the most well-known feature of the Saturnian system is its spectacular set of rings, seen in Fig. 21.1(b). Based on observations made from Earth, several fairly distinct rings have long been known to exist, labeled (from the outside in) A, B, and C. Between the prominent A and B rings, the **Cassini division** was thought to be virtually devoid of ring material. Another empty region, called the **Encke gap**, was observed within the A ring.

After the planet was visited by the Voyager spacecraft, other rings were discovered. The two spacecraft also revealed previously unexpected complexity in the system. As can be seen in Fig. 21.20(b), instead of large, almost continuous rings, thousands of *ringlets* were discovered; even the Cassini division has a number of rings lying in it, although the number density of the particles is much lower than in the neighboring regions. The F ring proved to be particularly perplexing because it is very narrow and appears to be braided [Fig. 21.21(a)].

The positions of the various rings and the Cassini division are given in Table 21.4; also included, from Example 19.2.1, is the estimate of Saturn's Roche limit for a satellite of density 1200 kg m^{-3} . The rings extend out as far as $8 R_S$ from Saturn, while the disk of the rings is very thin, perhaps only a few tens of meters thick. The presence of vertical ripples in the disk gives the rings the appearance of being about 1 km thick. Because of the thinness of the disk, when it is viewed perpendicular to the plane, the optical depth of the ring system ranges from about 0.1 to 2. It is actually possible to see through the rings in many locations.

When Galileo observed Saturn in 1612 (some two years after his initial observations), he was surprised to discover that the protrusions that he had seen earlier had apparently vanished! We now know that during his later set of observations, Galileo was viewing the rings edge-on, making them undetectable from Earth.

The reason the rings are so thin is easily understood by considering what happens to particles that undergo partially inelastic collisions, as shown in Fig. 21.22. Imagine two particles circling Saturn in the same direction but in orbits that are slightly tilted with respect to each other. If the two particles should collide, the x -components of their velocities would

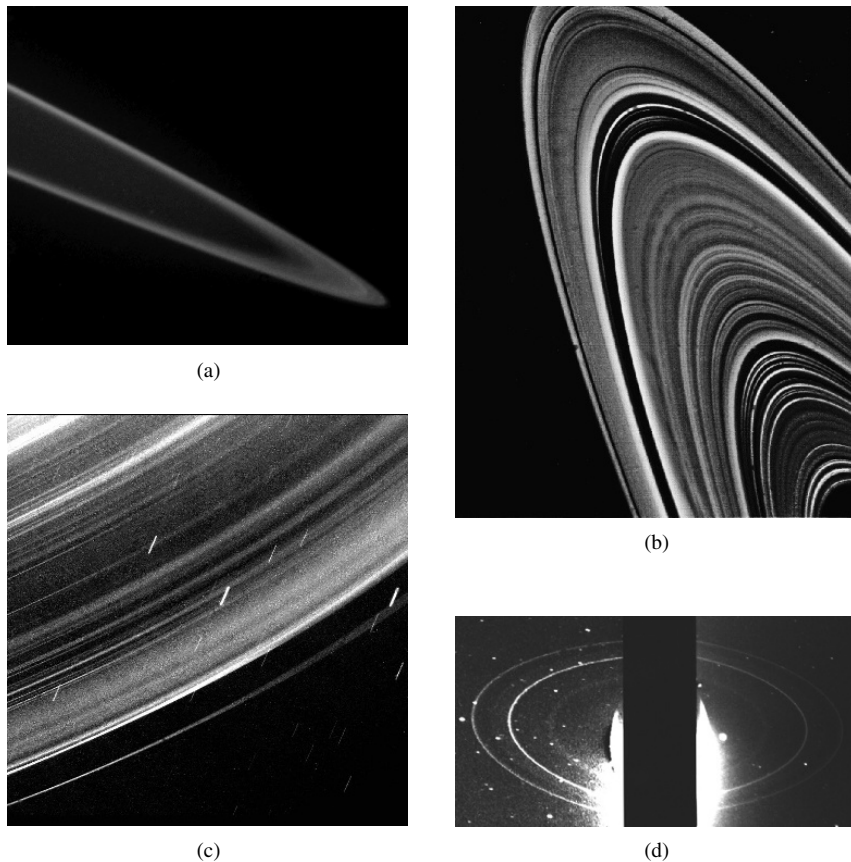


FIGURE 21.20 (a) Jupiter's very thin ring. (b) A close-up of Saturn's rings. Even the dark Cassini division is not entirely empty. (c) The rings of Uranus. Because the moving spacecraft was focused on the ring system, the background stars appear as streaks in the image. (d) Neptune was masked to bring out its faint ring system. (Courtesy of NASA/JPL.)

be largely unaffected, but the collision would decrease their y -components. The process diminishes the thickness of the disk until other effects start to become important, such as random collisions with incoming particles and perturbations from moons.

The Composition of Saturn's Rings

Most of the particles that make up the rings are quite small, with the majority having diameters that range from a few centimeters to several meters, although it seems likely that at least some particles with diameters as small as a few micrometers or as large as one kilometer may exist in the system. Size estimates are derived from several pieces of evidence, including the rate at which particles cool off in Saturn's shadow and how efficiently they reflect radar signals of various wavelengths.

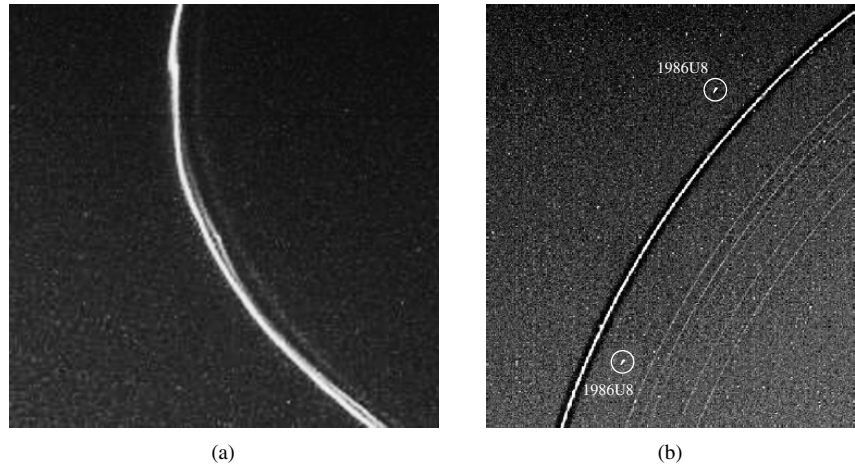


FIGURE 21.21 (a) Saturn's braided F ring. (b) Two small "shepherd" moons orbiting just inside and outside of the ϵ ring of Uranus. (Courtesy of NASA/JPL.)

TABLE 21.4 The Positions of Saturn's Ring Features.

Feature	Position (R_S)
D Ring	1.00–1.21
C Ring	1.21–1.53
B Ring	1.53–1.95
Cassini Division	1.95–2.03
A Ring	2.03–2.26
<i>Roche Limit</i>	2.04
F Ring	2.33
G Ring	2.8
E Ring	3–8

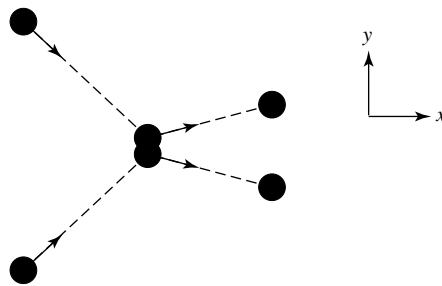


FIGURE 21.22 The collisions between particles help to keep the rings of Saturn very thin.

It has been known for some time that the material in the Saturnian rings is highly reflective (the rings have albedos in the range 0.2–0.6). Albedo measurements, combined with infrared spectroscopy, provide information about the composition of ring material. It appears that the particles of most of the rings are primarily water-ice, with some dust embedded in them or covering their surfaces. However, the very extended and thin E ring may be composed entirely of dust that is originating from the nearby moon, Enceladus.

Jupiter's Tenuous Ring System

Jupiter's very tenuous ring system [see Fig. 21.20(a)] has a characteristic optical depth of roughly 10^{-6} . Three components have been seen: the innermost *toroidal halo*, the *main* ring, and the outermost *gossamer* ring. Taken together they stretch from near Jupiter to about 3 R_J . It is believed that the ring material is primarily dust that is constantly resupplied by collisions of micrometeorites with larger objects (tiny moons) in the rings.

The Rings of Uranus

The rings of Uranus and Neptune [Figs. 21.20(c) and (d)] were first detected indirectly from Earth and later photographed by Voyager 2. On March 10, 1977, astronomers were observing the **occultation** of a background star (Uranus passing in front of the star). They were trying to measure the diameter of Uranus, as well as to gather some information about its atmosphere. Knowing the speed of the planet and the amount of time the star was hidden, they could determine the planet's diameter. Quite unexpectedly, the starlight faded and brightened several times before actually being occulted by the planet. When the star reappeared, the pattern of alternating fading and brightening was repeated, but in reverse order. The astronomers realized that rings were blocking out the star's light. The same procedure was used for Neptune but led to confusing results. In some cases the starlight was blocked on only one side of the planet, leading to the suggestion that only incomplete rings (or arcs) exist around Neptune.

A total of thirteen rings have been detected around Uranus, nine from the ground, two more by Voyager 2 in 1986, and another two by the Hubble Space Telescope in 2004. All of the rings are remarkably narrow, ranging in width from 10 km to 100 km (not unlike the F ring of Saturn), with some of the rings also showing signs of being braided. The two detected using the HST have diameters that are much greater than the other eleven rings, leading some researchers to refer to them as a second Uranian ring system.

The composition of the Uranian rings appears to be very different from that of either Jupiter or Saturn. Reflecting only about 1% of the incident sunlight, the ring material is extremely dark. This is because the rings are composed largely of dust, rather than ice.

Curiously, as mentioned in the previous section, the rings and moons of Uranus lie in the planet's equatorial plane and not along the ecliptic. Recalling that the rotation axis of Uranus is tilted 97.9° with respect to the ecliptic, this implies that the orientation of the orbits of the rings and moons changed after one or more catastrophic impacts dramatically shifted Uranus's axis (if indeed impacts were responsible). Apparently Uranus's rotationally produced equatorial bulge gravitationally affected its satellites and ring material, ultimately reorienting their orbits until the moons and rings were once again aligned with the planet's equator. Similarly, Saturn's rings are also aligned with its equator, despite the planet's equatorial plane being tilted almost 27° to its orbital plane.

Neptune's Rings

When it reached Neptune, Voyager 2 also found rings orbiting that planet. Like the rings of Uranus, several of the six identified rings are quite narrow, while the others appear to be diffuse sheets of dust. Oddly, the outermost ring, known as Adams,¹¹ has five discrete regions of concentrated material, like sausages on a string. It was these concentrations that were responsible for the arcs that were deduced from occultations.

Physical Processes Affecting Ring Systems

The spectacular observations of the Voyagers, Galileo, and Cassini have shown us that the dynamics of ring systems are quite complex. Not all of the features are as yet understood, but many important components have been identified:

- **Collisions** have already been mentioned as the process that maintains the thinness of the rings.
- **Keplerian shear** (or diffusion) spreads the rings out in the system's plane. As more rapidly moving particles in slightly lower orbits overtake more slowly moving particles farther out, collisions between them cause the inner particles to slow somewhat, and they drift closer to the planet. At the same time, the outer particles are accelerated, moving them outward. The process stops when the density of ring particles becomes so low that collisions effectively cease.
- **Shepherd moons** are small moons that reside in or near the edge of the rings, controlling the location of ring boundaries via their gravitational interactions. The narrowness of the F ring of Saturn [recall Fig. 21.21(a)] was understood when the two moons Pandora and Prometheus were discovered to be orbiting just outside and inside the ring, respectively. As the more rapidly moving ring particles pass Pandora, the moon's gravitational pull slows them down, causing the particles to drift inward. When Prometheus overtakes the ring particles, it pulls them forward, speeding them up and causing them to move outward. As a result, the F ring is confined to a narrow region just 100 km wide. Another shepherd moon (Atlas) defines the sharp outer edge of the A ring. Shepherd moons have also been discovered guiding one of the rings of Uranus; see Fig. 21.21(b).
- **Orbital resonances** between moons and ring particles in specific orbits can act to deplete or enhance particle concentrations. (It is also necessary that the orbital plane of the moon align with the ring plane.) For instance, there exists a 2:1 orbital resonance between Mimas and particles at the inner edge of the Cassini division. In other words, a particle in that location orbits twice for every orbit of Mimas. Since an inferior conjunction of such a particle with Mimas always occurs at the same position, gravitational perturbations of the particle's orbit produced by Mimas become cumulative, implying that the moon tends to force the particle into an elliptical orbit.

¹¹The Adams ring, along with the Leverrier and Galle rings, were named for the mathematical and observational discoverers of Neptune.

As the particle begins to cross the more circular orbits of particles at other radii, collisions become more likely. The outcome is that the particle has been removed from its original orbit and relocated in another part of the system (see Problem 21.13).

- **Spiral density waves**, first proposed by Peter Goldreich and Scott Tremaine in the late 1970s, are set up by moons as a consequence of orbital resonance. Gravitational perturbations can cause particles at different orbital radii to bunch up, effectively increasing their gravitational influence on other nearby particles in the disk. Those neighboring particles in turn are drawn toward the increase in density, extending the enhancement. If the moon responsible for the resonance is beyond the edge of the disk, the wave of density enhancement spirals outward. Since the density is larger in the wave, the probability of collision increases. Keplerian shear then causes the number density of particles near resonance orbits to decrease. This process helps explain the width of the Cassini division.¹²
- **The Poynting–Robertson effect** (a consequence of the headlight effect discussed in Example 4.3.3) can cause ring particles to spiral in toward the planet. When particles in the rings absorb sunlight, they must re-radiate that energy again if they are to remain in thermal equilibrium. The original light was emitted from the Sun isotropically, but in the Sun’s rest frame the re-radiated light is concentrated in the direction of motion of the particle. Since the re-radiated light carries away momentum as well as energy, the particle slows down and its orbit decays. This process is explored in more detail in Problems 21.15 and 21.16.
- **Plasma drag** is a consequence of the collisions of ring particles with charged particles trapped in the planet’s magnetic field. Since the magnetic field is anchored inside the planet, it must revolve with the rotation period of the planet. If the ring particles are inside the planet’s synchronous orbit (as most rings are), the particles will overtake the magnetic field plasma, and collisions will slow the particles down. The particles will then spiral in toward the planet, just as with the Poynting–Robertson effect. If the ring particles are outside the synchronous orbit, they will spiral outward.
- **Atmospheric drag** occurs as particles approach the outer reaches of the planet’s atmosphere. This effect quickly causes the particles to spiral down into the planet.
- **Radial spokes** have been observed in the rings of Saturn and are attributed to the interaction of charged dust particles with the planet’s magnetic field. These spokes move through the ring system with the rotation period of the planet, rather than with the orbital period of the rings. It appears that some dust particles acquire a net electrostatic charge as a result of their frequent collisions with other dust particles. This causes the dust to become trapped in magnetic field lines tens of meters above the plane of the rings. Sunlight that is scattered from the suspended particles produces the observed spokes.

¹²Spiral density waves also play an important role in the structure of spiral galaxies. They will be discussed in more detail in that context in Section 25.3.

- **Warping** of the disk is caused by the gravitational influences of the Sun and the planet's moons. If the Sun or the moons are not in exactly the same plane as the ring, particles in the ring are pulled out of the ring plane.

Ring Formation

The formation of planetary rings is still not fully understood. A major problem lies in the timescales involved in maintaining rings against processes that tend to disperse or destroy them. Are rings long-lived or transient phenomena? One idea, first suggested by Pierre-Simon Laplace (1749–1827) and Immanuel Kant (1724–1804) in the late 1700s, argues that rings are nebular in origin; they were formed at the same time that the planets accreted. Since most of Saturn's rings are composed largely of water-ice, while the rings of Jupiter, Uranus, and Neptune contain primarily nonvolatile substances (silicates and carbon), Saturn must have cooled more rapidly, before the water could escape. Although this idea could account for the spectacular Saturnian system, while also explaining the composition and greater sparsity of the rings of the other giant planets, it is difficult to understand how the systems could be maintained for more than 4.5 billion years.

It is also possible that ring systems arise due to tidal forces; if moons were drawn inside the planet's Roche limit, or if a comet or meteoroid ventured too close, tidal forces would fracture the objects, producing a new ring system. However, tidal disruption should leave intact rocky fragments as large as tens of kilometers in diameter. Grinding and meteoritic impacts would eventually break down the remnants, but such processes are extremely slow. On the other hand, loosely packed icy objects, such as comets, may be broken into smaller pieces by tidal disruptions (recall Shoemaker–Levy 9).

The discovery of the giant, outermost ring of Uranus by the HST was accompanied by the discovery of another moon, Mab, in the same orbit as the ring. It seems that when Mab is hit by meteorites, material ejected from the moon replenishes the giant ring, suggesting that the source for this ring, at least, has been identified.

Clearly, much work remains to be done before we can claim to understand the complexities of planetary rings.

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PROBLEMS

21.1 Estimate the pressures at the centers of Jupiter and Saturn. Compare your answers to the Sun’s central gas pressure.

21.2 Analytic functions can be derived for the pressure and density structure in the interior of Jupiter if an approximate relationship between pressure and density is assumed. A reasonable choice for a composition of pure molecular hydrogen is

$$P(r) = K\rho^2(r),$$

where K is a constant. This type of analytic model is known as a polytrope (see page 334ff for a discussion of polytropic stellar models).

- (a) By substituting the expression for the pressure into the hydrostatic equilibrium equation (Eq. 10.6) and differentiating, show that a second-order differential equation for the density can be obtained, namely

$$\frac{d^2\rho}{dr^2} + \frac{2}{r} \frac{d\rho}{dr} + \left(\frac{2\pi G}{K} \right) \rho = 0.$$

- (b) Show that the equation is satisfied by

$$\rho(r) = \rho_c \left(\frac{\sin kr}{kr} \right),$$

where ρ_c is the density at the center of the planet and

$$k \equiv \left(\frac{2\pi G}{K} \right)^{1/2}.$$

- (c) Taking the average radius of Jupiter to be $R_J = 6.99 \times 10^7 \text{ m}$ and assuming that the density goes to zero at the surface (i.e., $kR_J = \pi$), determine the values of k and K .
- (d) Integrate Eq. (10.7) using the analytical solution for Jupiter's density as a function of radius to find an expression for the planet's interior mass, M_r , written in terms of r and ρ_c . *Hint:*

$$\int r(\sin kr) dr = \frac{1}{k^2} \sin kr - \frac{r}{k} \cos kr.$$

- (e) Using the boundary condition that $M_r = M_J$ at the surface, estimate the planet's central density. (The value obtained in this problem is lower than the result found from detailed numerical calculations, 1500 kg m^{-3} . One major reason for the difference is that Jupiter's composition is not purely molecular hydrogen.)
- (f) Make separate plots of the density and interior mass as functions of radius.
- (g) What is the central pressure of your model of Jupiter? (One detailed model gives a value of $8 \times 10^{12} \text{ N m}^{-2}$.)
- 21.3** (a) Assuming spherical symmetry and the density distribution for the polytropic model of Jupiter given in Problem 21.2b, show that the moment of inertia is given by

$$I = \frac{8\rho_c}{3\pi} \left(1 - \frac{6}{\pi^2} \right) R_J^5.$$

Hint: Since this analytical model does not assume constant density, you will need to integrate over concentric rings to find the moment of inertia about Jupiter's rotation axis. Recall that $I \equiv \int_{\text{vol}} a^2 dm$, where $a = r \sin \theta$ is the distance from the rotation axis to the ring of mass dm , and

$$dm = \rho(r) dV = \rho(r) 2\pi ar d\theta dr.$$

- (b) Using the estimated value for the central density of Jupiter obtained in Problem 21.2e, calculate the planet's moment-of-inertia ratio. *Hint:* The moment-of-inertia ratio was first introduced in Problem 20.12.
- (c) Compare your answer to part (b) with the measured value given in Table 21.2. What does this result say about the true density distribution within the planet relative to the analytical model?

- 21.4 (a)** Show that Eq. (21.5) follows directly from Eq. (21.3) in cylindrical coordinates for the case of a two-component model planet with two constant densities (see Fig. 21.5). Assume that the outer component is oblate and the inner component is spherical.
- (b)** Verify that Eq. (21.5) reduces to the familiar case of

$$I_{\text{sphere}} = \frac{2}{5}MR^2$$

for a spherically symmetric planet of constant density.

- 21.5 (a)** Derive an equation for the mass of the core of the two-component planetary model shown in Fig. 21.5. You should express your answer in terms of the fractional equatorial radius, fR_e , and the constant density of the core.
- (b)** Assume that Jupiter has a $10 M_{\oplus}$ core and that the average density of the core is $15,000 \text{ kg m}^{-3}$. Determine f , the ratio of the equatorial radius of the planet's core to the equatorial radius of its surface.
- (c)** What is the average envelope density in this two-component model?
- (d)** Determine the moment-of-inertia ratio (I/MR_e^2) for this two-component model.
- (e)** Compare your answer in part (d) to the measured value of the moment-of-inertia ratio for Jupiter given in Table 21.2. What can you say about the mass distribution of Jupiter compared to the analytical model?
- 21.6** Estimate the angular diameter of Jupiter's magnetosphere as viewed from Earth at opposition. Compare your answer with the angular diameter of the full Moon.
- 21.7** Suppose that fragment G of comet Shoemaker–Levy 9 measured 700 m in diameter. If this fragment had an average density of 200 kg m^{-3} , estimate its kinetic energy just before it entered the planet's atmosphere. You may assume that it struck the atmosphere with a speed equal to the planet's escape speed. Express your answer in joules and megatons of TNT ($1 \text{ Mton} = 4.2 \times 10^{15} \text{ J}$).
- 21.8 (a)** On the *same scale*, plot (1) the first-order correction term to the gravitational potential of Saturn as a function of θ [i.e., the J_2 term in Eq. (21.1)], (2) the second-order correction term, and (3) the sum of the two terms. (Similar plots for Jupiter are shown in Fig. 21.4; note that the plots for Jupiter use different scales.) Assume that the observer is a distance $r = 2R_e$ from the planet.
- (b)** For which angle(s) is the gravitational potential largest? smallest? By what percent do these values for the gravitational potential deviate from the case of spherical symmetry (the zeroth-order term)?
- 21.9 (a)** Estimate the amount of energy radiated by Jupiter over the last 4.55 billion years (see Eq. 10.23).
- (b)** Estimate the *rate* of energy output from Jupiter due to gravitational collapse alone, assuming that the rate has been constant over its lifetime.
- (c)** Compare your answer for part (b) with the value for the intrinsic power output that was given in the text. What does this say about the rate of energy output in the past? Discuss the implications for the evolution of the Galilean moons.
- 21.10** Estimate the blackbody temperature of Neptune, taking into consideration that one-half of all the energy radiated by the planet is due to internal energy sources. Compare your answer with the measured value of $59.3 \pm 1.0 \text{ K}$.

- 21.11** Assume that all of the ions escaping Io are sulfur ions. Assuming also that this rate has been constant over the last 4.55 billion years, estimate the amount of mass lost from the moon since its formation. Compare your answer with Io's present mass (8.932×10^{22} kg).
- 21.12** (a) Make a rough estimate of the mass contained in Saturn's rings. Assume that the rings have a constant mass density and that the disk is 30 m thick with an inner radius of $1.5 R_S$ and an outer radius of $3 R_S$ (neglect the E ring). Assume also that all of the ring particles are water-ice spheres of radius 1 cm and that the optical depth of the disk is unity. The density of the particles is approximately 1000 kg m^{-3} . *Hint:* Refer to Eq. (9.12) to estimate the number density of water-ice spheres.
- (b) If all of the material in Saturn's rings were contained in a sphere having an average density of 1000 kg m^{-3} , what would the radius of the sphere be? For comparison, the radius and mass of Mimas are 196 km and 4.55×10^{19} kg, respectively, and it has an average density of 1440 kg m^{-3} .
- 21.13** Carefully sketch the orbits of Mimas and a characteristic Saturnian ring particle that is locked in a 2:1 orbital resonance with the moon. Show qualitatively that the resonance produces an elliptical orbit.
- 21.14** Calculate the position of Saturn's synchronous orbit. Are any of its rings located outside of that radius? If so, which ones?
- 21.15** A dust grain orbiting the Sun (or in a planetary ring system) absorbs and then re-emits solar radiation. Since the light is radiated from the Sun isotropically and re-emitted by the grain preferentially in the direction of motion, the particle is decelerated (it loses angular momentum) and spirals in toward the object it is orbiting. This process (known as the Poynting–Robertson effect) is just a consequence of the headlight effect discussed in Example 4.3.3.
- (a) If a dust grain orbiting the Sun absorbs 100% of the energy that strikes it and all of the energy is then re-radiated so that thermal equilibrium is maintained, what is the luminosity of the grain? Assume that the particle's cross-sectional area is σ_g and its distance from the Sun is r .
- (b) Show that the *rate* at which angular momentum is lost from a grain is given by

$$\frac{d\mathcal{L}}{dt} = -\frac{\sigma_g}{4\pi r^2} \frac{L_\odot}{mc^2} \mathcal{L}, \quad (21.6)$$

where m and $\mathcal{L} = mvr$ are the mass and angular momentum of the grain, respectively, and $L_\odot$ is the luminosity of the Sun. *Hint:* Think of radiated photons as carrying an *effective mass* away from the grain; the effective mass of a photon is just $m_\gamma = E_\gamma/c^2$.

- 21.16** (a) Beginning with Eq. (21.6), show that the time required for a spherical particle of radius R and density ρ to spiral into Saturn from an initial orbital radius R_0 is given by

$$t_{\text{Saturn}} = \frac{8\pi\rho c^2}{3L_\odot} R r_S^2 \ln\left(\frac{R_0}{R_S}\right),$$

where R_S is the radius of the planet and r_S is its distance from the Sun. Assume that the orbit of the particle is approximately circular at all times and that it is always a constant distance from the Sun.

- (b) The E ring is known to contain dust particles having average radii of $1 \mu\text{m}$. If the density of the particles is 3000 kg m^{-3} , how long would it take for a typical particle to spiral into the planet from an initial distance of $5 R_S$?

- (c) Compare your answer in part (b) to the estimated age of the Solar System. Could the E ring be a permanent feature of the Saturnian system without a source to replenish the ring? Note that the small moon Enceladus orbits Saturn in the E ring.

21.17 The mass and radius of Miranda are 8×10^{19} kg and 236 km, respectively.

- (a) What is the escape velocity from the surface of Miranda?
- (b) What would be the speed of a small object freely falling toward Uranus when it crossed the orbit of Miranda? Assume that the object started falling toward Uranus from rest, infinitely far from the planet. Neglect any effects due to the orbital motion of the planet around the Sun. Miranda's orbital radius is 1.299×10^8 m.
- (c) Using Eq. (10.22), estimate the amount of energy needed to pulverize that moon.
- (d) Suppose that a spherical object with a density of 2000 kg m^{-3} were to collide with Miranda, completely destroying it. If the object hit Miranda with the speed found in part (b), what would the object's radius need to be? *Note:* For the purposes of this “back-of-the-envelope” calculation, you need not be concerned with the energy that would be expended in pulverizing the impacting object.

CHAPTER

22

Minor Bodies of the Solar System

- 22.1 *Pluto and Charon*
- 22.2 *Comets and Kuiper Belt Objects*
- 22.3 *Asteroids*
- 22.4 *Meteorites*

22.1 ■ PLUTO AND CHARON

The success of the mathematical prediction of Neptune's position (page 775) led astronomers to consider the possibility that a ninth planet existed even farther from the Sun. Based on *perceived* anomalies in the orbits of Uranus and Neptune, the search began in the late nineteenth century. Finally, on February 18, 1930, after a systematic and tedious search, Clyde W. Tombaugh (1906–1997) discovered a small 15th-magnitude object orbiting the Sun. The new object was classified as a planet and named Pluto for the Roman god of the underworld.¹ It turns out that even though Pluto was discovered near its predicted position, the prediction was invalid because it was founded on statistically insignificant apparent deviations in the orbits of the other planets.

Pluto is unlike any of the terrestrial or giant planets of the Solar System; in fact, it bears much more resemblance to Neptune's moon Triton than to any of the planets discussed in Chapter 20 or 21. Its 248.5-year orbit is very eccentric ($e = 0.25$). At perihelion it is only 29.7 AU from the Sun (actually closer than Neptune), while it is 49.3 AU away at aphelion. Its orbit is also inclined significantly from the ecliptic (17°).

Despite the fact that Pluto is a Neptune-crossing object, it is not in any danger of colliding with the giant world. Pluto is protected from that fate by a 3:2 orbital resonance with Neptune. Consequently, Pluto is never near perihelion when it is in conjunction with Neptune, and the two planets never get any closer than about 17 AU. Pluto actually approaches Uranus more closely, coming within 11 AU.

The Discovery of Charon

Many of the most basic characteristics of Pluto, such as its mass and radius, were poorly determined until its largest moon, **Charon**, was discovered in 1978.² Figure 22.1 shows a

¹The name Pluto was suggested by Venetia Burney, who was then an 11-year-old English schoolgirl.

²Prior to the discovery of Charon, Pluto's radius was uncertain to within a factor of 4, and its mass wasn't known to better than a factor of 100.

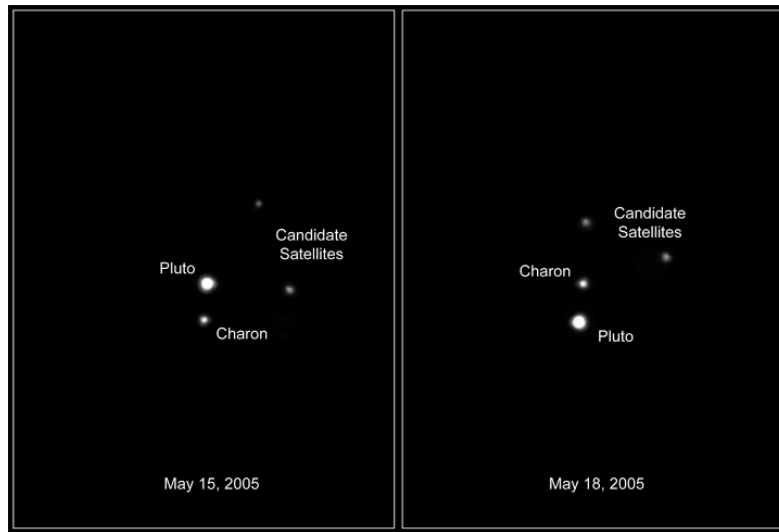


FIGURE 22.1 Pluto and its three moons. Charon was discovered in 1978, and the other two moons were detected by the Hubble Space Telescope's Advanced Camera for Surveys in 2005. [Courtesy of NASA, ESA, H. Weaver (JHU/APL), A. Stern (SwRI), and the Hubble Space Telescope Pluto Companion Search Team.]

Hubble Space Telescope image of the Pluto system. Pluto and Charon orbit the system's center of mass in 6.39 d with a separation of only 1.964×10^7 m (just slightly more than $1/20$ of the distance between Earth and the Moon). Using Kepler's third law, the combined mass of the system is only $0.00247 M_{\oplus}$. Of course, to determine each mass individually requires knowledge of the ratio of their separations from the system's center of mass, which yields a mass ratio of $M_{\text{Charon}}/M_{\text{Pluto}} = 0.124$. From these data, the mass of Pluto has been estimated to be 1.3×10^{22} kg, and Charon's mass is roughly 1.6×10^{21} kg; for comparison, Triton's mass is 2.14×10^{22} kg.

The Densities and Compositions of Pluto and Charon

Shortly after the discovery of Charon, astronomers realized that a rare **eclipse season** would occur between 1985 and 1990. Since the orbital plane of the Pluto–Charon system is inclined 122.5° to their orbit around the Sun, observers on Earth see the system edge-on only for brief intervals once every 124 years. Fortuitously, Pluto was also at perihelion in 1989. The next eclipse season will not occur until the twenty-second century.

The duration of the occultations provided the information necessary to determine the radius of Charon. Pluto's radius has been determined to be 1137 km, making it only about two-thirds the size of our Moon, but the occultation data indicate that Charon's radius is about 600 km. This means that Pluto's average density is about 2110 kg m^{-3} and Charon's density is roughly 1770 kg m^{-3} . These data seem to indicate that Pluto and Charon are probably made of frozen ices and rock, with Pluto having a somewhat higher proportion of rock than the majority of the moons of the giant planets. The best map of the surface of Pluto obtained to date is shown in Fig. 22.2.

It is worth noting that Triton's density is 2050 kg m^{-3} , very similar to that of Pluto.

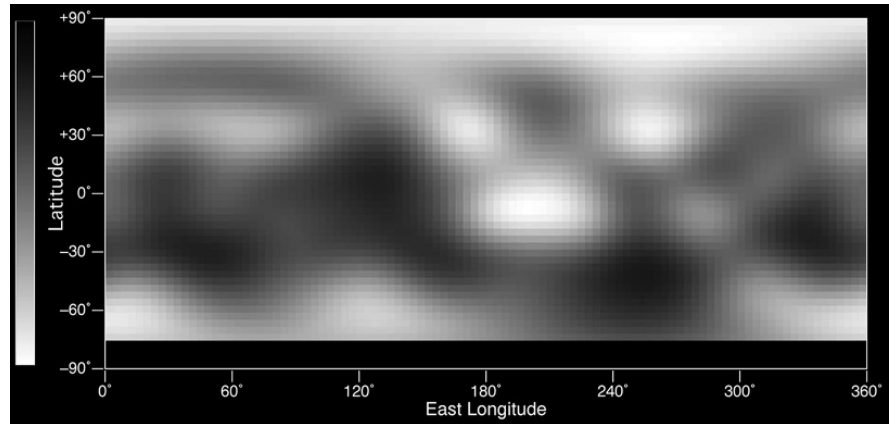


FIGURE 22.2 The surface of Pluto as seen by the Hubble Space Telescope in 1994. 85% of the surface is represented; the south polar region was pointed away from Earth when the map was constructed. The observed features may be basins and craters, or perhaps frost from various ices. [Courtesy of Alan Stern (Southwest Research Institute), Marc Buie (Lowell Observatory), NASA, and ESA.]

The Possible Formation of Charon by a Large Impact

With a mass that is almost $1/8$ that of Pluto, Charon is proportionately the most massive moon in the Solar System relative to its parent planet.³ It appears that Charon must have formed as a result of a large impact on Pluto, similar to the way scientists believe our Moon formed around Earth (recall Fig. 20.20). It is also possible that the two moons discovered in 2005 are additional products of that collision. The impactor probably had a mass of between 0.2 and $1 M_{\text{Pluto}}$.

Complete Spin–Orbit Coupling

Pluto and Charon also have another interesting dynamical characteristic, briefly mentioned in Section 19.2: Both objects have rotation periods that are exactly the same as their orbital period about their mutual center of mass. Since they spin in the same direction as their orbital motion, Pluto and Charon keep the same faces toward each other at all times; they are completely locked in a synchronous orbit. The tidal forces between these two small worlds have resulted in the final state of lowest energy. Because they are fully locked, the tidal forces do not produce the constantly changing bulges seen in other systems, such as Earth's tidal bulges produced by the Moon. Therefore, the frictional heat losses and angular momentum transfer in operation elsewhere have now ceased for the Pluto–Charon interaction.

A necessary consequence of the locked, synchronous orbit is that Charon is located directly over Pluto's equator. If this weren't the case, the orbital motion would carry Charon alternately north and south of Pluto's equator, and any deviation from spherical symmetry between the two worlds would result in constantly changing tidal forces. Since the orbital

³Our Moon, which is the second-largest satellite in the Solar System relative to its parent planet, contains only about $1/81$ the mass of Earth.

plane of the system is inclined 122.5° to its orbit around the Sun, this implies that both Pluto and Charon rotate retrograde. Recall that Uranus also rotates retrograde and that its ring system and regular satellites are located directly over its equator; that orientation has also been attributed to tidal forces.

A Frozen Surface and a Changing Atmosphere

In 1992 Tobias C. Owen and collaborators used the United Kingdom Infrared Telescope on Mauna Kea, Hawaii, to carry out a spectroscopic study of Pluto's surface. Their work revealed that the surface is covered with frozen nitrogen (N_2), which constitutes some 97% of the total area, with carbon monoxide ice (CO) and methane ice (CH_4) each accounting for 1% to 2%, similar to the surface of Triton. Oddly, Charon's surface appears to be composed primarily of water-ice; no molecular nitrogen, carbon monoxide, or methane ices or gases have been detected on Charon.

When Pluto occulted a faint star in 1988, a very tenuous atmosphere was detected, with a surface pressure of about 10^{-5} atm. The atmosphere is dominated by N_2 , with CH_4 and CO probably making up roughly 0.2% of the total by number, consistent with the composition of the surface ice and the rate of sublimation of the various species. Curiously, when Pluto occulted another star in 2002, measurements of the pressure and scale height of the atmosphere had doubled, implying that Pluto's atmosphere had become significantly thicker over the 14-year period.

It has been suggested that the atmosphere of this tiny, distant world is not permanent. The 1988 observation of its atmosphere was made near perihelion when the planet's temperature was near its maximum value of approximately 40 K. At this temperature, ices on the surface are able to undergo partial sublimation. The atmosphere apparently thickened between 1988 and 2002 because ices continued to sublimate, releasing additional gases into the atmosphere. However, as the planet moves back toward aphelion, the atmosphere will probably "freeze out" again.

A Rendezvous with Pluto

In January 2006, NASA launched the **New Horizons** flyby mission. If all goes according to plan, New Horizons will pass near Pluto in 2015, giving us the first close-up look at this tiny, distant planet and its moons. Whether Pluto will still have an atmosphere at the time of the flyby is one of the many interesting questions waiting to be answered about this Trans-Neptunian quadruple system.

22.2 ■ COMETS AND KUIPER BELT OBJECTS

Comets such as Comet Mrkos (Fig. 11.25) and Comet Halley [Fig. 22.3(a)], have been observed frequently throughout history. In fact, the periodic visits of Comet Halley have been recorded during each passage through the inner Solar System since at least 240 B.C. (its orbital period is 76 years).⁴ Because of their unusual appearance when near perihelion, bright

⁴This famous comet has probably been making its periodic tours of the inner Solar System for 23,000 years, but it was Edmond Halley who first realized that these were repeated observations of the same object. The proof of his hypothesis was accomplished with the aid of Newton and his newly developed mechanics; see Section 2.2.

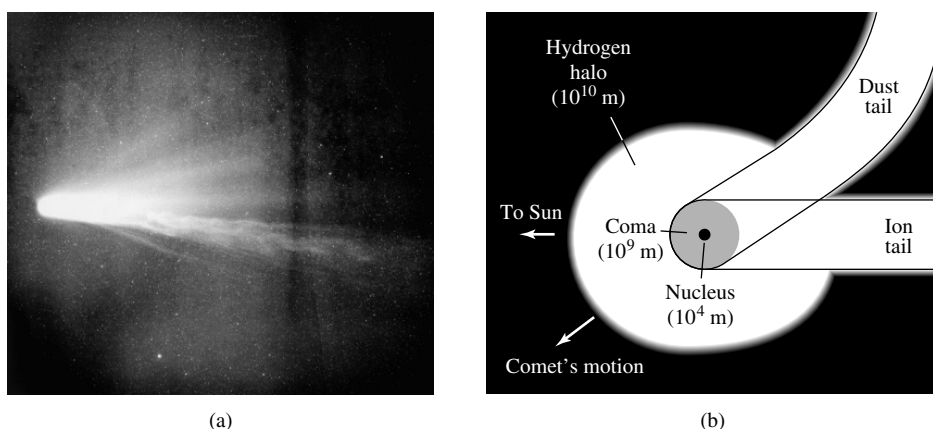


FIGURE 22.3 (a) Comet Halley clearly exhibited both a dust tail (curved) and an ion tail (straight) during its most recent trip through the inner Solar System. This image was obtained by the Michigan Schmidt telescope at Cerro Tololo Interamerican Observatory on April 12, 1986. Visible is a detachment event occurring in the ion tail. (Courtesy of NASA/JPL.) (b) The anatomy of a comet.

comets have long been associated with mystery and forces beyond human comprehension. While many people believed that comets foretold coming evils, others considered them messengers of good news. The artist Giotto di Bondone (1266–1337) depicted the “Star of Bethlehem” as a comet in his work *Adoration of the Magi*, which adorns the interior of the Scrovegni Chapel in Padua, Italy (Fig. 22.4). The painting dates from 1303, just two years after Comet Halley had last appeared.

A Model of a Comet

In 1950 Fred L. Whipple (1906–2004) proposed a model of comets that successfully explained most of their physical characteristics, including the development of **tails** when they enter the inner Solar System; see Fig. 22.3(b). He suggested that a “dirty snowball,” roughly 10 km across, lies at the center of the comet. This dirty snowball constitutes the comet’s **nucleus**. The nucleus is composed of ices and embedded dust grains. As the nucleus moves from the cold environment of the outer Solar System into the warmer regions near the Sun, the ices begin to sublimate. The released dust and gas then expand outward to produce a 10^9 -m cloud of gas and dust known as the **coma**. Subsequently, the material in the coma interacts with the sunlight and solar wind, producing the very long, familiar tails (up to 1 AU in length) that are always associated with our images of comets. We now know that a hydrogen gas **halo** (or envelope) also surrounds the coma and can have a diameter of 10^{10} m. When the comet leaves the inner Solar System, the temperature decreases sufficiently that the rate of sublimation diminishes significantly and the halo, coma, and tails disappear. Cometary activity does not stop entirely, however; small outbursts can occur from time to time as heat from the Sun travels inward through the nucleus, releasing highly volatile gases.



FIGURE 22.4 *Adoration of the Magi*, by Giotto di Bondone (1266–1337). The painting adorns the interior of the Scrovegni Chapel in Padua, Italy. Work on the painting began two years after the 1301 appearance of Comet Halley.

The Dynamics of Comet Tails

Comet tails are always directed away from the Sun, as depicted in Fig. 22.5. Two independent mechanisms are responsible for the structure of the tails: radiation pressure on the liberated dust grains, and the interaction of ions with the solar wind and the Sun's magnetic field.

First consider the effect of radiation pressure on grains. For an *idealized* spherical dust grain of radius R that is located a distance r from the Sun and that absorbs all of the incident light that strikes it, Eq. (3.13) can be used to calculate the outward force of radiation pressure on the grain. The factor of $\cos \theta$ means that the grain's cross-sectional area $\sigma = \pi R^2$ should be used in calculating the force. In other words, for the case of complete absorption the same force would be exerted if the grain were replaced by a circular disk with an identical radius, oriented perpendicular to the light. Using $\langle S \rangle = L_\odot / 4\pi r^2$ for the magnitude of the time-averaged Poynting vector, the force on the grain due to radiation pressure is

$$F_{\text{rad}} = \frac{\langle S \rangle \sigma}{c} = \frac{L_\odot}{4\pi r^2} \frac{\pi R^2}{c}. \quad (22.1)$$

Of course the Sun's gravity is also acting on the grain. If the density of the grain is ρ , its mass is

$$m_{\text{grain}} = \frac{4}{3} \pi R^3 \rho$$

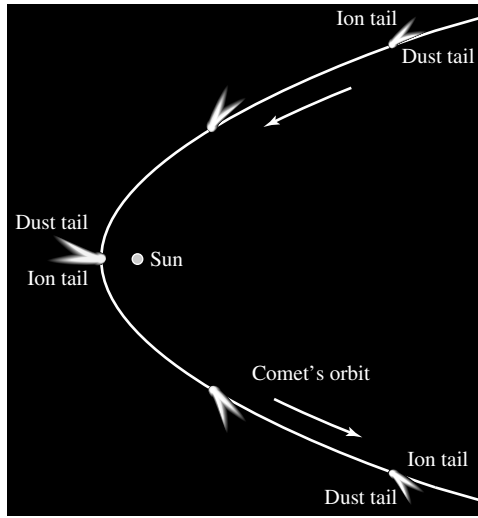


FIGURE 22.5 The curved dust tail and the straight ion tail are always directed away from the Sun.

and the magnitude of the gravitational force acting on it is given by

$$F_g = \frac{GM_\odot m_{\text{grain}}}{r^2} = \frac{4\pi GM_\odot \rho R^3}{3r^2},$$

where F_g is directed inward toward the star. Now, the ratio of the magnitudes of these forces is

$$\frac{F_g}{F_{\text{rad}}} = \frac{16\pi GM_\odot R \rho c}{3L_\odot}. \quad (22.2)$$

Because both gravity and light obey an inverse square law, this ratio does not depend on the distance r from the Sun.

The magnitudes of the gravitational and radiation pressure forces on a cometary dust grain will be equal if the radius R of the grain has the critical value

$$R_{\text{crit}} = \frac{3L_\odot}{16\pi GM_\odot \rho c}. \quad (22.3)$$

Grains *smaller* than R_{crit} will experience a net outward force and will spiral away from the Sun. The curvature of the dust tail arises because of the decrease in the orbital speeds of the grains with increasing distance from the Sun. For a typical density of $\rho = 3000 \text{ kg m}^{-3}$, the critical radius of the dust grain is found to be $R_{\text{crit}} = 1.91 \times 10^{-7} \text{ m} = 191 \text{ nm}$.

Grains larger than R_{crit} will continue to orbit the Sun. However, a competing process, the Poynting–Robertson effect, causes larger grains to spiral slowly in toward the Sun. As will be shown in Problem 22.2, the time required for a spherical particle of radius R and

density ρ to spiral into the Sun from an initial orbital radius r is given by⁵

$$t_{\text{Sun}} = \frac{4\pi\rho c^2}{3L_{\odot}} Rr^2. \quad (22.4)$$

Because R_{crit} is comparable to the wavelengths λ of light emitted by the star, the actual situation is more complicated than this simple analysis suggests. The smallest grains ($R \ll \lambda$) are inefficient absorbers of light and have absorption cross sections that are much smaller than πR^2 . Furthermore, the dust grains will scatter some of the incident light rather than absorb it. The effect of scattering on the radiation pressure force depends on the composition and geometry of the dust grains and on the wavelength of the light.

The Compositions of Comets

It is the scattering of light that gives dust tails their white or yellowish appearance. On the other hand, the color of an ion tail is blue because CO^+ ions absorb and reradiate solar photons at wavelengths near 420 nm. However, CO^+ is certainly not the only species identified in comets. To date, a rich variety of atoms, molecules, and ions have been discovered spectroscopically, including some molecules that are quite complex. During its 1986 perihelion passage, the inner coma of Comet Halley was found to contain roughly (by number) 80% H_2O , 10% CO , 3.5% CO_2 , a few percent $(\text{H}_2\text{CO})_n$ (polymerized formaldehyde), 1% methanol (CH_3OH), and traces of other compounds. A partial list of chemical species that have been found in comets is given in Table 22.1.

Disconnection Events

The straight ion (or plasma) tail owes its structure to a complex interaction among the comet, the solar wind, and the Sun's magnetic field.⁶ Since the comet is an obstruction in the path of the solar wind and because the relative speed of the solar wind and the comet exceeds both the local sound speed (Eq. 10.84) and the Alfvén speed (Eq. 11.11), a shock front develops in the direction of motion.⁷ As matter piles up at this *bow shock*, ions in the coma get trapped in the Sun's magnetic field, loading the field down. This causes the magnetic field to wrap around the nucleus. The cometary ions circle the field lines and trail behind the nucleus in the antisolar direction. A **disconnection event** occurs when the comet encounters a reversal in the solar magnetic field (the boundary between magnetic field directions is known as a *sector boundary*). During a disconnection event, the ion tail breaks away and a new one forms in its place. Disconnection events are quite common and are evident in the image of Halley's comet in Fig. 22.3(a), as well as in the time sequence of Comet Hyakutake in March 1996 (Fig. 22.6).

Robotic Investigations of Comets

Whipple's dirty-snowball hypothesis was dramatically verified in March 1986, when an international armada of spacecraft rendezvoused with Comet Halley. The fleet was made up of

⁵Recall that the Poynting–Robertson effect is also important in understanding the dynamics of Saturn's rings.

⁶The model of a cometary *magnetotail* was first proposed by Hannes Alfvén in 1957.

⁷Shocks were first discussed in Section 11.2.

TABLE 22.1 A Partial List of Chemical Species Found in Comets. Various isotopomers have also been detected, such as HDO (deuterium replacing one of the hydrogen atoms in H₂O).

Atoms	Molecules	Ions
H	CH	H ⁺
C	C ₂	C ⁺
O	CN	Ca ⁺
Na	CO	CH ⁺
Mg	CS	CN ⁺
Al	NH	CO ⁺
Si	N ₂	N ₂ ⁺
S	OH	OH ⁺
K	S ₂	H ₂ O ⁺
Ca	H ₂ O	H ₂ S ⁺
Ti	HCH	CO ₂ ⁺
V	HCN	H ₃ O ⁺
Cr	HCO	H ₃ S ⁺
Mn	NH ₂	CH ₃ OH ₂ ⁺
Fe	C ₃	
Co	OCS	
Ni	H ₂ CO	
Cu	H ₂ CS	
	NH ₃	
	NH ₄	
	CH ₃ OH	
	CH ₃ CN	
	(H ₂ CO) _n	

two spacecraft from Japan (**Suisei** and **Sakigake**), two from the former Soviet Union (**Vega 1** and **Vega 2**),⁸ one from the European Space Agency (**Giotto**, named for the twelfth-century Italian painter; recall page 817), and one from the United States (the **International Cometary Explorer**, which flew past Comet Giacobini–Zinner six months earlier).⁹ A number of spacecraft that were already flying other missions temporarily trained their instruments on the comet as well, even though they did not actually go out and meet the famous visitor.

On March 6 and 9, from distances of 8900 km and 8000 km, respectively, Vega 1 and Vega 2 were able to obtain low-resolution pictures of the nucleus of the comet. Relaying the Vega telemetry data to the European Space Agency, the Giotto scientists were then able to guide their spacecraft to an even closer encounter. As a direct result of extraordinary

⁸The two Soviet missions traveled to Venus first, releasing probes into its atmosphere.

⁹ICE was originally named the **International Sun–Earth Explorer 3** when it was launched in 1978. After many years on another mission, the spacecraft was reassigned to investigate the two comets.

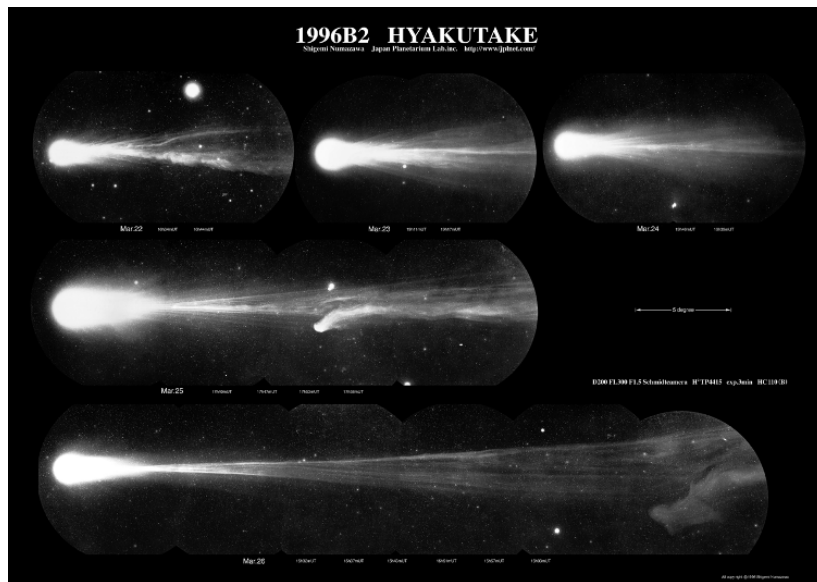


FIGURE 22.6 A spectacular example of a disconnection event in the March 25 image of this sequence of images of Comet Hyakutake in 1996. (©Shigemi Numazawa/Atlas Photo Bank/Photo Researchers, Inc.)

international cooperation, Giotto came within 596 km of the nucleus on March 14 (the mission goal was 540 km).¹⁰

Because of the high relative velocity of Giotto and Halley (68.4 km s^{-1}), even a collision with a small dust grain from the nucleus could cause serious damage to the spacecraft. Therefore, to protect the instruments on board, Giotto was equipped with a 50-kg shield made of aluminum and plastic Kevlar. Despite these precautions, just seven seconds before its closest approach to the nucleus, a dust particle struck the spacecraft off-axis, causing it to wobble severely. Stability was restored again one-half hour later.

Near its closest approach, the Giotto camera recorded the image of the nucleus seen in Fig. 22.7. The size of Whipple's dirty snowball is approximately $15 \text{ km} \times 7.2 \text{ km} \times 7.2 \text{ km}$, and it is shaped roughly like a potato. The surface is extremely dark, with an albedo of between 0.02 and 0.04. Repeated trips near the Sun have apparently left behind a layer of dust and possibly organic material as the ices evaporated away.

Visible in Fig. 22.7 are dust jets (ejected streams of material) located on the Sunward side of the nucleus. The positions of the jets probably correspond to thin regions in the dark covering on the surface through which trapped, heated gases in the interior are able to escape. As the nucleus rotates, other surfaces are exposed to the Sun, and new jets develop.¹¹

¹⁰“The Halley Encounters,” by Rüdiger Reinhard, in *The New Solar System*, Third Edition, Beatty and Chaikin (eds.), Cambridge University Press and Sky Publishing Corporation, Cambridge, MA, 1990, pp. 207–216, offers a fascinating recounting of the flybys.

¹¹The nucleus of Comet Halley appears to have two rotation periods (2.2 days and 7.4 days) corresponding to motions about different axes of this irregularly shaped object.



FIGURE 22.7 The nucleus of Comet Halley as seen by the Giotto spacecraft. Evident are jets of gases on the sunward side of the nucleus. (Image from Reitsema et al., *20th ESLAB Symp.*, ESA SP-250, Vol. II, 351, 1986.)

It has been estimated that during the flybys, jets were present on 15% of the surface, with gas and dust discharge rates of approximately $2 \times 10^4 \text{ kg s}^{-1}$ and $5 \times 10^3 \text{ kg s}^{-1}$, respectively. Because of the reaction forces produced by the jets, comets tend to have slightly erratic orbits.

By considering these nongravitational perturbations to its orbit, the mass of Comet Halley's nucleus has been estimated at between $5 \times 10^{13} \text{ kg}$ and 10^{14} kg . If these crude estimates turn out to be correct, the average density of the nucleus is less than 1000 kg m^{-3} and may be as low as 100 kg m^{-3} . It is likely that gases and dust escaping from the icy body have left a porous, honeycomb structure inside the nucleus with an average density close to that of new-fallen snow. (Recall that the Shoemaker–Levy 9 fragments that hit Jupiter also appear to have been very loosely packed, with densities of about 600 kg m^{-3} ; see Section 21.1.)

The passage of the International Cometary Explorer through the tail of Comet Giacobini–Zinner and the extensive international collaboration to rendezvous with Comet Halley were the first two of several missions to comets that have occurred since the mid-1980s. In 2001 the experimental ion-propulsion spacecraft **Deep Space 1** obtained images of the 10-km-long nucleus of Comet Borrelly from a vantage point of 2200 km. In addition, on January 2, 2004, **Stardust** passed within 250 km of Comet Wild 2, obtaining high-resolution images of the surface. Stardust also captured dust from the comet and, on January 15, 2006, returned the dust to Earth for analysis.

A very dramatic encounter occurred on July 4, 2005, when **Deep Impact** sent a 370-kg impactor into Comet Tempel 1, a 7.6 km by 4.9 km object, at a speed of 10.2 km s^{-1} . Images of Comet Tempel 1 obtained by Deep Impact are shown in Fig. 22.8. The impactor created

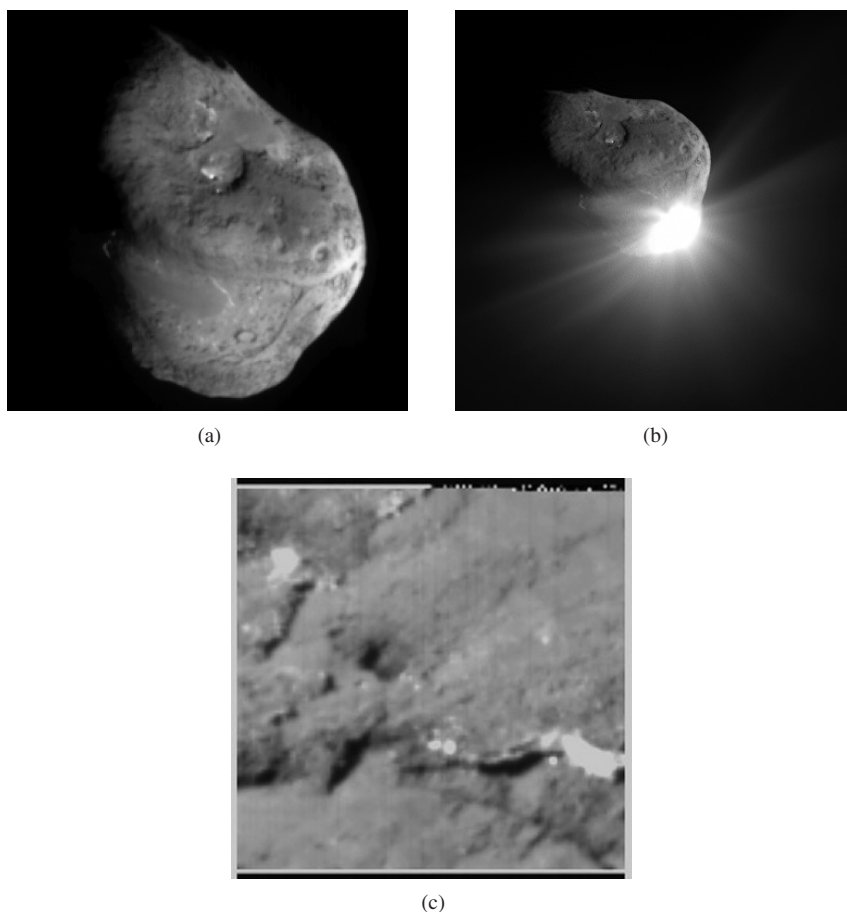


FIGURE 22.8 (a) Comet Tempel 1 seen from Deep Impact five minutes before the impactor struck the comet. The nucleus measures 5 km by 11 km. (b) 67 seconds after impact. (c) The surface of Tempel 1, 20 seconds before impact. The image was obtained by the impactor's targeting sensor. Features as small as 4 m can be identified. (NASA/JPL-Caltech/UMD.)

a crater in the comet's nucleus that allowed scientists to study the interior. Analysis of the data identified a variety of compounds, including water, carbon dioxide, hydrogen cyanide, methyl cyanide, polycyclic aromatic hydrocarbons (PAHs), and other organic molecules, as well as minerals such as olivine, calcite, iron sulfite, and aluminum oxide. The 10^7 kg of material excavated from the near surface of the comet has the consistency of a very fine sand, or perhaps even the consistency of talcum powder. It also seems that the comet's density is roughly that of powder snow, loosely held together by its weak gravity. With a mass of 7.2×10^{13} kg, Tempel 1 has a density of only 600 kg m^{-3} . The low density suggests that Tempel 1 may be a very porous rubble pile.

In the future, ESA's **Rosetta** spacecraft, which was launched in 2004, is scheduled to rendezvous with Comet Churyumov–Gerasimenko in 2014 and then spend almost two years

carefully studying the comet from a low orbit around the nucleus. The mission will also land a probe on the surface of the comet. During its extended tour, Rosetta will fly alongside the comet as it moves into the inner Solar System, is heated by the Sun, and begins releasing trapped volatiles from its interior. Among other goals, this exhaustive study may help us to determine whether or not the “seeding” of Earth with cometary organic material may have helped life develop on our planet.

Sun-Grazing Comets

While many comets that venture into the inner Solar System suffer relatively small amounts of sublimation near perihelion compared with their masses, other comets may experience more severe consequences. For instance, when Comet West moved through the inner Solar System in 1976, its nucleus broke into four separate pieces. Comet Kohoutek also split apart in 1974. Perhaps more impressive are the **Sun-grazing comets**. While it has been intensely studying the Sun, the LASCO instrument onboard the SOHO spacecraft has discovered more than 1000 comets that make close approaches to the Sun. In some instances the comets’ orbits cause them to plunge into the Sun, as occurred with Comet SOHO-6, shown in Fig. 22.9.

The Oort Cloud

Halley is one example of a class of comets known as **short-period comets** with orbital periods of less than 200 years. Short-period comets are found near the ecliptic and return to the inner Solar System repeatedly. The **long-period comets** have orbital periods of greater than 200 years, and some may take from 100,000 to 1 million years or more to return. In 1950, based on a very careful statistical study of their apparently random orbits, Jan Oort (1900–1992) concluded that long-period comets originate in a distant distribution of cometary nuclei now known as the **Oort cloud**. Although the Oort cloud has never been

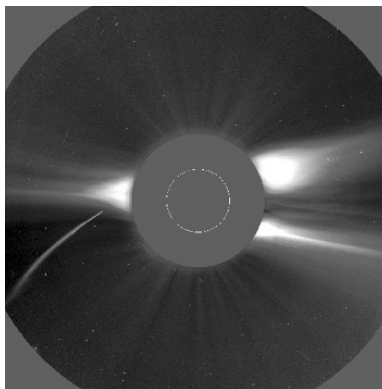


FIGURE 22.9 An image of Comet SOHO-6 plunging into the Sun (lower left-hand side). The disk of the Sun is covered so that the Sun’s corona and coronal mass ejections can be studied. The image was obtained by the LASCO instrument onboard SOHO on December 23, 1996. [SOHO (ESA & NASA).]

observed, its existence seems certain. As we noted in Section 19.1, the reservoir of nuclei appears to be located between 3000 AU and 100,000 AU from the Sun and probably contains 10^{12} – 10^{13} members with a total mass on the order of $100 M_{\oplus}$. For comparison, the nearest stars are approximately 275,000 AU away. The inner Oort cloud (3000 AU to 20,000 AU) may be slightly concentrated along the ecliptic, while the outer Oort cloud (20,000 AU to 100,000 AU) has a nearly spherical distribution of cometary nuclei.

The comets in the Oort cloud probably did not form at their present locations. Instead, they may be ancient planetesimals that coalesced near the ecliptic in the vicinity of Uranus and Neptune. After repeated gravitational interactions with the ice giants, the nuclei were catapulted out to their current distances. Because these nuclei were so far from the Sun, passing stars and gas clouds ultimately randomized their orbits, resulting in the nearly spherical distribution that exists in the outer cloud today. With the inner cloud being deeper in the Sun's gravitational well, these cometary nuclei did not become quite as randomly distributed. Therefore, comets in the inner Oort cloud were able to retain some history of their original locations when the Solar System was young. It is probably the gravitational perturbations of other stars and gas clouds that cause some of the cometary nuclei to start their long falls into the inner Solar System.

The Kuiper Belt

Since the orbits of short-period comets lie preferentially near the ecliptic, it seems unlikely that these objects originated in the Oort cloud. Kenneth E. Edgeworth (1880–1972) in 1949 and Kuiper in 1951 independently proposed that a second collection of cometary nuclei might be located close to the plane of the ecliptic.

In August 1992, 1992 QB₁, a 23rd-magnitude object, was discovered by Jane Luu and David Jewitt 44 AU from the Sun and having an orbital period of 289 years. Seven months later a second 23rd-magnitude object (1993 FW) was discovered at nearly the same distance from the Sun. Assuming these objects have albedos characteristic of typical cometary nuclei (3% to 4%), then they must have diameters of approximately 200 km in order to appear as bright as they do. That would make them about one-tenth the size of Pluto. By early 2006, telescopic surveys employing sensitive CCD cameras had resulted in the discovery of more than 900 similar objects beyond Neptune's orbit.

Now known as the **Kuiper belt**,¹² this disk of cometary nuclei extends from 30 to 50 AU from the Sun; the semimajor axis of Neptune's orbit is 30 AU. Some members appear to have particularly eccentric orbits that may reach out to 1000 AU at aphelion, however. Noting their location beyond the outermost ice giant, these **Kuiper Belt Objects** (KBOs) are sometimes alternatively referred to as **Trans-Neptunian Objects** (TNOs).

A Kuiper Belt Object Larger Than Pluto

As more and more KBOs have been discovered, a number of them have been found to have diameters somewhat smaller than, but comparable to, that of Pluto (see Table 22.2). However, in 2005, astronomers Mike Brown (Caltech), Chad Trujillo (Gemini Observatory), and David Rabinowitz (Yale University) announced the discovery of the first object in the

¹²In recognition of the independent suggestion of Edgeworth, this collection of objects is sometimes referred to as the **Edgeworth–Kuiper belt**.

TABLE 22.2 A List of the Largest Known Kuiper Belt Objects, as of May 2006. Many of the listed diameters are quite uncertain.

Name	Diameter (km)	Period (yr)	a (AU)	e	i (deg)
2003 UB313	2400	559	67.89	0.4378	43.99
Pluto	2274	248	39.48	0.2488	17.16
Sedna*	1600	12,300	531.7	0.857	11.93
Orcus	1500	247	39.39	0.220	20.6
Charon	1270	248	39.48	0.2488	17.16
2005 FY9	1250	309	45.71	0.155	29.0
2003 EL61	1200	285	43.34	0.189	28.2
Quaoar	1200	287	43.55	0.035	8.0
Ixion	1070	249	39.62	0.241	19.6
Varuna	900	282	42.95	0.052	17.2
2002 AW197	890	326	47.37	0.131	24.4

* Sedna has an orbit that is much larger than the classical Kuiper belt.

Kuiper belt known to be larger than Pluto. 2003 UB313 was discovered on January 5, 2005, from data that were collected in a sky survey in 2003. With an orbital period of $P = 559$ yr, a semimajor axis of $a = 68$ AU, an orbital eccentricity of $e = 0.44$, and an inclination with respect to the ecliptic of $i = 44^\circ$, the orbital characteristics of 2003 UB313 are why it took so long to find this large object in the Kuiper belt. Since most surveys looking for KBOs have been focused near the plane of the ecliptic, it was surprising to find 2003 UB313 with such a large inclination. HST observations indicate that the diameter of 2003 UB313 is 2400 km, making it about 6% larger than Pluto. The spectrum of 2003 UB313, shown in Fig. 22.10, is also strikingly similar to Pluto's, suggesting a surface composition dominated by frozen methane. A moon has been detected orbiting 2003 UB313 as well.

The discovery of 2003 UB313 has rekindled the debate over what officially constitutes a planet. Should 2003 UB313 be designated as a planet since it is larger than Pluto? Should Pluto be removed from the planet classification? Interestingly, prior to the discovery of large KBOs, no formal scientific definition of a planet had ever been established; it had more or less been assumed that we will recognize a planet when we see one! At the time of writing, this is an issue that the International Astronomical Union is struggling to resolve. Given that society has long come to see Pluto as a planet, the decision may very well be beyond the formal definitions of scientists. Whatever the resolution of these questions, it is clear that in light of the apparent size and composition of 2003 UB313 and many of the other KBOs, Pluto and Charon are certainly Kuiper belt objects; they just happen to be among the largest known members. After all, Pluto and Charon have more in common with very large cometary nuclei than with a "typical" planet-moon system. Neptune's unusual moon, Triton, is probably a captured KBO as well.

Classes of Kuiper Belt Objects

As the number of known KBOs has continued to grow, it has become evident that they fall into three different groups based on their orbital characteristics. **Classical KBOs** are

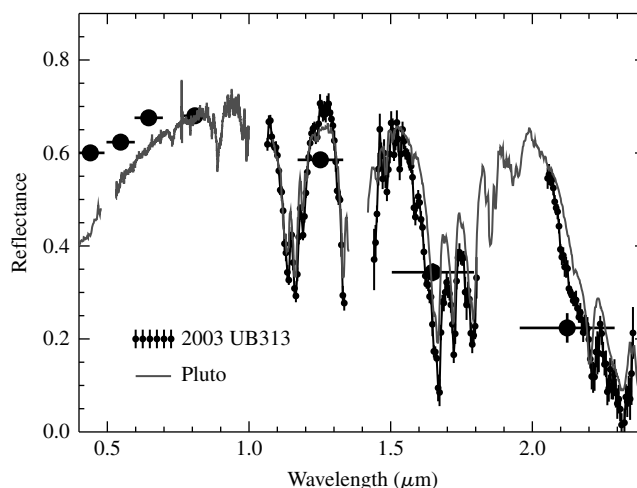


FIGURE 22.10 The reflection spectrum of 2003 UB313 (individual points) compared to the spectrum of Pluto (gray line). Absorption features of methane dominate the spectrum. The large points are data from *BVR/IRHK* photometry. [Courtesy of Mike Brown (Caltech), Chad Trujillo (Gemini Observatory), and David Rabinowitz (Yale University).]

those that orbit between 30 and 50 AU from the Sun, most have semimajor axes between 42 and 48 AU. The orbital inclinations of classical KBOs tend to be less than 30° . It has been suggested that the cutoff of classical KBOs at 50 AU may be due to a passing star early in the formation of the Solar System. **Scattered KBOs** have much higher orbital eccentricities than the classical KBOs and were probably pumped up to those orbits by gravitational interactions with the ice giants, most notably Neptune. 2003 UB313 is one example of a scattered KBO. The perihelion distances of scattered KBOs are characteristically about 35 AU, and they tend to have greater orbital inclinations than the classical KBOs. In addition, it is likely that the scattered KBOs are at least one source of the short-period comets. Finally, a class of **resonant KBOs** exists that have orbital resonances with Neptune. As we noted in Section 22.1, Pluto is locked in a 3:2 orbital resonance with Neptune that protects it from ever colliding with the ice giant. As a result, Pluto (and Charon) are resonant KBOs. In fact, KBOs that have 3:2 orbital resonances with Neptune are referred to as **Plutinos**. Orbital resonances of 4:3, 5:3, and 2:1 have been observed for KBOs as well.

Centaurs

Other fairly large icy bodies have been discovered orbiting the Sun as well. In 1977, an object known as 2060 Chiron was detected in an orbit that carries it from inside Saturn's orbit out to the orbit of Uranus. Chiron's diameter has been estimated to be between 200 km and 370 km. A slightly smaller object, nicknamed "Son of Chiron" but officially called 5145 Pholus, has also been seen orbiting among the outer planets, between 8.7 and 32 AU. Originally classified as an asteroid, Chiron brightened unexpectedly in 1988 and developed a measurable coma. Such behavior defines this object as a comet rather than a rocky asteroid. Chiron and 5145 Pholus are two examples of a class of objects known as **Centaurs** that

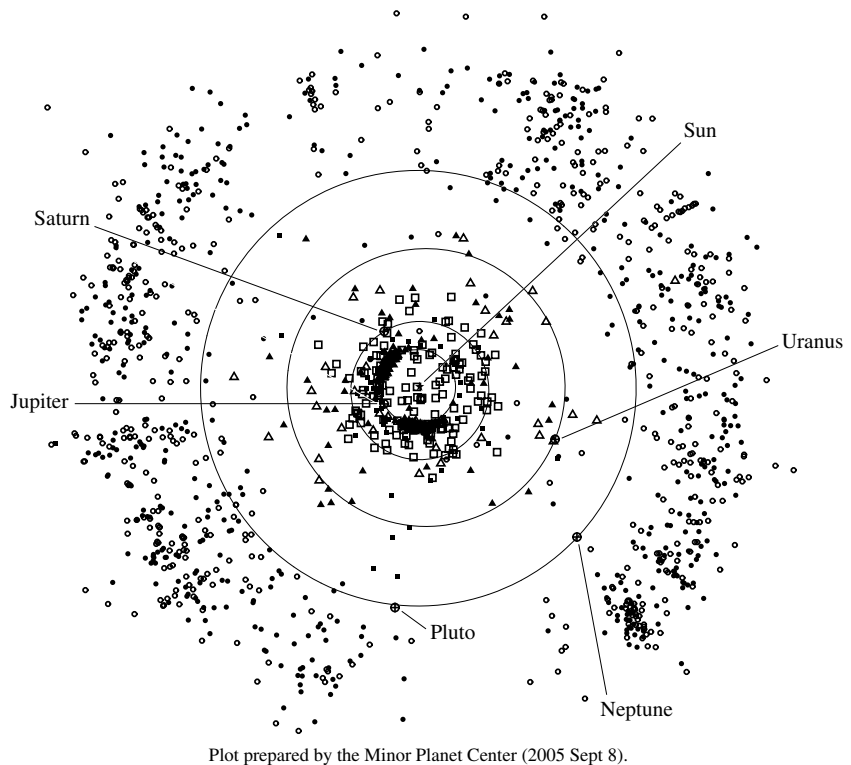


FIGURE 22.11 The positions of known classical and resonant Kuiper belt objects (circles), Centaurs and scattered KBOs (triangles), comets (squares), and Trojan asteroids from the orbit of Jupiter outward on September 8, 2005. The position of each object has been projected onto the plane of the ecliptic. Open symbols are objects observed at only one opposition; filled symbols are objects that have been observed at multiple oppositions. For the comets, filled squares represent numbered periodic comets; other comets are indicated by open squares. The classical Kuiper belt is clearly evident beyond the orbit of Neptune. In this orientation, the planets and most of the other objects orbit counterclockwise, and the vernal equinox is to the right. (Adapted from a figure courtesy of Gareth Williams, Minor Planet Center.)

appear to be KBOs which were scattered into the region of the planetary orbits. Centaurs may eventually become short-period comets.

Figure 22.11 shows the locations of KBOs, Centaurs, comets, and Jupiter's Trojan asteroids (see Section 22.3) on September 8, 2005, projected onto the plane of the ecliptic. Because of the projection effect and the very large sizes of the symbols relative to the actual sizes of the objects, the outer Solar System appears to be more crowded than it really is.

The Implications for Water in the Inner Solar System

As we shall see in Section 23.2, it seems unlikely that the terrestrial planets could have condensed out of the warm inner solar nebula with large abundances of volatiles such as

water. It has been suggested that much of the water that is found in the oceans of Earth, that is trapped in the permafrost and ice caps of Mars, and that probably existed on Venus in the past could have been delivered to those worlds by impacting comets after the planets formed. However, “the devil is in the details.” After careful examination of the composition of several comets that have been explored by spacecraft, it has been noted that the deuterium-to-hydrogen (D/H) ratio in the comets exceeds that in Earth’s oceans by at least a factor of two. In fact, the D/H ratio is more characteristic of the interstellar medium than of the terrestrial oceans. From the small sample of comets that have been closely investigated to date, it appears that another source for the water in Earth’s oceans must be determined. On the other hand, it may be that the sample is biased, consisting only of objects that probably derived from the Oort cloud, rather than the Kuiper belt. Of course, it is also possible that the delivery of water to Earth was a prolonged process, involving a variety of mechanisms, including comets, asteroids with relatively high water content (Section 22.3), water-rich meteorites (Section 22.4), and planetesimals (Section 23.2).

22.3 ■ ASTEROIDS

As was already mentioned in Section 19.1, **asteroids** (sometimes referred to as **minor planets**) usually occupy orbits that are closer to the Sun than most comets. The vast majority of asteroids can be found in a belt situated between the orbits of Mars and Jupiter. Since the discovery of Ceres in 1801, several hundred thousand asteroids have been cataloged, and it may be that the total number is in excess of 10^7 . However, despite their large numbers, the combined mass of all the asteroids may be as low as $5 \times 10^{-4} M_{\oplus}$. A close-up view of asteroid 243 Ida and its moon, Dactyl, is shown in Fig. 22.12. (The number designates the order in which the asteroid was discovered. Ceres is designated 1 Ceres.)

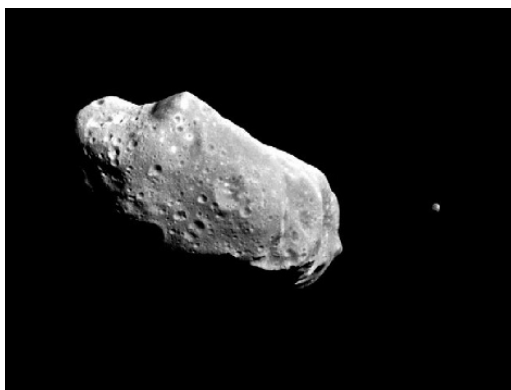


FIGURE 22.12 243 Ida and its moon, Dactyl, as seen by the Galileo spacecraft on August 28, 1993, during its journey to Jupiter. 243 Ida is 55 km long, and Dactyl (100 km from Ida at the time of the flyby) is somewhat egg-shaped, measuring 1.6 km by 1.2 km. Surface features as small as 30 m are visible on Ida. Galileo was about 10,500 km from Ida when this image was taken. (Courtesy of NASA/JPL.)